

Architectural Blueprint: Enterprise Real Estate Development Intelligence and Automated Feasibility Platform

Reverse Engineering and Source Truth Analysis

The traditional real estate land acquisition and pre-development evaluation lifecycle is characterized by fragmentation and latency. In conventional practice, evaluating whether a prospective urban land parcel is physically buildable, legally compliant, and financially viable requires between four and six weeks of uncoordinated effort across disparate disciplines. Municipal development control regulations—such as the Development Control and Promotion Regulations for Greater Mumbai (DCPR 2034) or the Unified Development Control and Promotion Regulations for Maharashtra State (UDCPR)—are manually parsed from dense legal gazettes. Cadastral property perimeters are traced from paper revenue survey sheets or isolated CAD files, and financial pro-formas are calculated in static spreadsheets disconnected from spatial geometry.

The reference materials, consisting of the *Planwise Site Development Masterclass* and the *Land Feasibility Platform Blueprint*, document an interactive browser prototype developed to accelerate early-stage site assessment. The platform links client-side WebGL canvas manipulation with spatial deductions, municipal constraint forms, and basic real-time metrics. However, an architectural audit reveals fundamental limitations in the reference implementation: calculations occur largely within the client browser, geographic coordinates are processed in non-conformal projections, geometric offsetting relies on basic approximations, and downstream architectural layout engines are referenced only conceptually.

The enterprise platform specified herein transforms this early concept into a production-grade Real Estate Development Intelligence Platform. The system executes a deterministic computational pipeline: ingesting cadastral land parcels, projecting them into conformal metric spaces, applying declarative statutory bye-laws, executing exact straight skeleton geometric offsets, synthesizing generative site plans, packing architectural floor plates adhering to statutory building codes, solving multi-level parking layouts, extruding 3D massings, and running institutional discounted cash flow underwriting and multi-objective Pareto optimizations.

To maintain technical integrity, every architectural assertion, functional capability, and technical constraint is classified under five explicit epistemic tiers:

- [DOCUMENTED]: Directly supported by the uploaded masterclass or blueprint documents.
- [INFERRED]: Reasonably inferred through standard geospatial and software engineering patterns.
- [EXTERNALLY VERIFIED]: Supported by authoritative statutory planning codes, building bye-laws, or computational geometry literature.
- [PROPOSED]: New architectural designs, optimized algorithms, and scalable patterns designed to transcend prototype limitations.
- [UNKNOWN]: System details omitted from the primary source texts with no determinable evidence.

Planwise Source Truth Matrix

The following matrix establishes the empirical baseline extracted from the reverse-engineering audit, identifying documented capabilities, inferred structures, critical unknowns, and proposed enterprise equivalents.

Functional Capability	Source Evidence	Epistemic Status	Technical Confidence	Unknowns and Undocumented Areas	Proposed Modern Enterprise Equivalent
Product Purpose	Compresses land acquisition feasibility from 4–6 weeks to minutes by digitizing boundaries and evaluating spatial constraints.	[DOCUMENTED]	High	Integration pathways to municipal automated building approval portals (e.g., AutoDCR).	End-to-end automated regulatory evaluation, parametric massing, and open BIM/CAD export pipeline.
Target Users	Real estate developers, master planners, architects, land aggregators, and transaction advisory firms.	[DOCUMENTED]	High	Multi-tenant enterprise organization hierarchies, team permission boundaries, and client-sharing models.	Multi-tenant RBAC with custom profiles (Owner, Architect, Engineer, Financial Analyst, Auditor).
User Journey	Sequential lifecycle: Authentication → Dashboard → Project Setup → Boundary Delineation → Feature Placement → Constraint Entry →	[DOCUMENTED]	High	State machine handling for branched project revisions, phasing, or multi-parcel assemblies.	Directed Acyclic Graph (DAG) workflow orchestrator supporting design branching and scenario tracking.

Functional Capability	Source Evidence	Epistemic Status	Technical Confidence	Unknowns and Undocumented Areas	Proposed Modern Enterprise Equivalent
	Summary Review.				
Frontend Architecture	Single Page Application (SPA), React 18, functional components, hooks (useState, useRef, useCallback, useEffect), React Router v6.	[DOCUMENTED]	High	Global state management configuration (Zustand, Redux) or server-state caching mechanisms.	React 19 / TypeScript, TanStack Query v5, Zustand client store, Zod runtime schema validation.
Map Engine	Mapbox GL JS coupled with @mapbox/mapbox-gl-draw, custom SVG controls, and Mapbox Geocoder restricted to India (countries: 'in').	[DOCUMENTED]	High	Custom WebGL shader implementations for shadow casting and terrain draping; projection handling.	MapLibre GL JS / Mapbox GL JS v3 with custom WebGL Three.js layers running on a shared GPU context.
GeoJSON Handling	RFC 7946 WGS84 coordinates for boundaries (Polygon), deductions (Polygon), and trees (Point); stringified API serialization.	[DOCUMENTED]	High	Mathematical projection transformations from EPSG:4326 degrees to local planar metric coordinates.	Native JSONB PostGIS persistence with automated topological validation (ST_MakeValid) and UTM projection.
Authentication Flow	Email/password submission returning signed JWT; token persisted in browser localStorage; injected into	[DOCUMENTED]	High	Refresh token rotation, multi-factor authentication, and server-side session invalidation.	OAuth2 / OIDC gateway with ephemeral in-memory access tokens and encrypted HttpOnly, SameSite=Stric

Functional Capability	Source Evidence	Epistemic Status	Technical Confidence	Unknowns and Undocumented Areas	Proposed Modern Enterprise Equivalent
	Axios headers.				t cookies.
Database Tier	Relational schemas referenced; Supabase and PostgreSQL mentioned; API listening on port 5001.	[INFERRED]	Medium	Exact DBMS engine, spatial indexes (GiST), table schemas, foreign keys, and migration scripts.	PostgreSQL 16 with PostGIS 3.4, Prisma/SQLAlchemy migrations, PgBouncer pooling, and multi-tenant RLS.
Auto-Save Mechanism	1000ms debounced auto-save hook using useRef timer and AbortController cancellation for PUT mutations.	[DOCUMENTED]	High	Distributed conflict resolution when multiple collaborators concurrently edit identical parcels.	Operational Transformation (OT) or Yjs CRDTs over WebSockets paired with debounced durable writes.
Toolbar State Machine	Dynamic toolbar disabling tree/deduction tools until boundary polygon exists via toolActivationRules.	[DOCUMENTED]	High	Precision CAD drafting tools (snapping, orthogonal locking, vertex fillets, chamfers, edge splitting).	Formal XState finite state machine supporting precision CAD drawing, object snapping, and history undo/redo.
Spatial Geometry	Geodesic area calculation, ring closure verification, and basic point-in-polygon containment checks.	[DOCUMENTED]	Medium	Inward setback offsetting, straight skeleton generation, Minkowski sums, and boolean polygon clipping.	Dedicated C++ / Python geometry computation service integrating CGAL, GEOS, and Clipper2.
Regulatory Model	Manual form entry for abutting road width,	[DOCUMENTED]	High	Dynamic rule engine capable of compiling and evaluating	Declarative, version-controlled JSON/DSL Rule Engine

Functional Capability	Source Evidence	Epistemic Status	Technical Confidence	Unknowns and Undocumented Areas	Proposed Modern Enterprise Equivalent
	front/rear/side setbacks, height limits, and land ownership enums.			contradictory statutory legal clauses.	compiling municipal statutes (DCPR, UDCPR, JBBL).
Feasibility Workflow	Net plot area derivation: gross plot area minus road reservations, non-developable cutouts, and tree buffers.	[DOCUMENTED]	High	Algorithmic resolution of buildable envelopes on non-convex, acute-angled, or fragmented plots.	Fully deterministic Feasibility Pipeline resolving setbacks, gross built-up area (BUA), and statutory FSI caps.
Generative Layout	Downstream layout tools mentioned for column grids, corridors, and 1BHK/2BHK/3BHK units.	[UNKNOWN]	Low	Algorithmic implementation of generative layout synthesis (MILP vs. Wave Function Collapse).	Constrained Mixed-Integer Linear Programming (MILP) layout synthesizer adhering to NBC 2016 norms.
3D Massing	Extrusion of 2D floor plates into 3D structural volumes across podiums, transfer slabs, and tower envelopes.	[INFERRED]	Low	Procedural 3D mesh generation, step-back offsets, tower tapering, and shadow casting.	Procedural Three.js parametric massing engine with real-time shadow projection and solar exposure analysis.
Financial Engine	Mentions FinWise integration; high-level IRR, NPV, and Residual Land Value (RLV) calculations.	[INFERRED]	Low	Institutional underwriting equations, multi-phase cash flows, debt servicing, and waterfall schedules.	Deterministic real estate financial engine modeling GDV, TDC, approval costs, TDR purchases, and IRR/XIRR.
Reporting	Basic summary	[DOCUMENTED]	Medium	Vector drawing	Headless

Functional Capability	Source Evidence	Epistemic Status	Technical Confidence	Unknowns and Undocumented Areas	Proposed Modern Enterprise Equivalent
Engine	review card (InputSummary) and PDF download triggers.	[DOCUMENTED]		generation, dynamic pro-forma tables, and statutory calculation audit trails.	Chromium / PDFKit compilation pipeline streaming multi-page investment memorandums.
Testing Strategy	Manual debugging using Chrome DevTools, console inspection, and network payload verification.	[DOCUMENTED]	High	Automated unit tests, geospatial fuzzing suites, visual regression, and CI/CD pipelines.	Multi-tier test harness: Vitest, Hypothesis property-based testing, PostGIS regression tests, Playwright E2E.
Deployment Setup	Local development environment running on localhost:3000 (React SPA) and localhost:5001 (API service).	[DOCUMENTED]	High	Cloud infrastructure, Kubernetes manifests, container definitions, and CDN caching.	Kubernetes-orchestrated container cluster (AWS EKS), Redis, Kafka/RabbitMQ, and S3-compatible storage.
Security Model	Route guarding via <ProtectedRoute> reading localStorage JWT; vulnerable to cross-site scripting (XSS).	[DOCUMENTED]	Medium	Enterprise security: HttpOnly cookies, CSRF protection, CSP directives, and encryption.	Enterprise security perimeter: HttpOnly dual-token rotation, PostgreSQL RLS, TLS 1.3, strict CSP headers.

Technical Product Scope and Functional Architecture

The platform provides a comprehensive system architecture comprising twenty-seven modular capabilities designed to operate with complete mathematical determinism and strict data isolation.

The authentication subsystem enforces multi-tenant identity lifecycle management via an

OAuth2 and OIDC compliant gateway, supporting SAML 2.0 federation for institutional enterprise accounts, hardware-token multi-factor authentication, and cryptographically signed session tokens. The organizational hierarchy establishes tenant boundaries that partition data, compute resources, seat licensing, and administrative privileges. Within each organization, projects serve as version-controlled containers holding design briefs, parcel boundary iterations, and regulatory parameters.

The cadastral management module coordinates the ingestion of municipal revenue surveys, land records, and parcel boundary definitions, allowing planners to merge multiple titles or subdivide larger tracts. The geospatial vector map renders satellite basemaps, terrain elevation models, and custom vector layers on the GPU at sixty frames per second. The spatial context layer executes spatial joins against municipal infrastructure databases, incorporating zoning districts, coastal regulation zones, flight paths, and utility rights-of-way. Precision CAD drawing tools allow planners to draft and modify geometries with vertex, edge, and midpoint snapping, orthogonal angle tracking, and dynamic dimension inputs.

The computational geometry engine serves as the platform's core mathematical processing layer, executing polygon validation, straight skeleton offsetting, Minkowski dilations, and boolean clipping in local projected metric coordinates. The declarative regulation engine isolates statutory planning codes from application logic, compiling municipal bye-laws into structured rule definitions. The constraint engine evaluates physical and legal restrictions across parcels, executing geometric subtractions for road widenings, environmental corridors, and tree protection buffers. The feasibility engine resolves statutory dependencies deterministically, calculating net plot areas, base and incentive Floor Space Index (FSI), and maximum permissible building heights.

The generative site planning engine synthesizes macro-scale development configurations, optimizing building footprints, open spaces, and vehicular access routes. The floor-plate layout engine proceduralizes interior architectural configurations, synthesizing structural cores, egress corridors, and MEP service shafts. The residential unit packing module applies mixed-integer linear programming to optimize unit mixes while enforcing statutory room dimensions, window placements, and structural bay alignments. The mobility engine calculates vehicular parking stall counts, circulation aisles, access ramps, and fire tender turning clearances across subterranean basements, ground stilts, and multi-level podiums. The structural grid engine assists early-stage planning by generating regular column networks (6.0m to 7.5m spans) and flagging vertical misalignments between residential towers and parking grids that require structural transfer slabs.

The 3D massing engine converts 2D floor plates into interactive 3D models rendered within the WebGL viewport using a single, shared graphics context. The financial feasibility engine evaluates project cash flows, modeling land costs, municipal approval fees, hard construction costs, financing interest, and sales milestones to derive Net Present Value (NPV), Internal Rate of Return (IRR), and Residual Land Value (RLV). The scenario comparison engine tracks alternative development configurations across design branches, supporting comparative analysis of financial yields, structural complexities, and unit mix variations. The multi-objective optimization engine uses an evolutionary genetic algorithm (NSGA-II) to balance competing design priorities, mapping a Pareto frontier of viable development configurations.

The AI development copilot operates through a natural language interface, translating user queries into structured tool invocations. The copilot operates behind strict deterministic validation gates: it can query and explain municipal regulations, but cannot invent or modify geometric, regulatory, or financial values. The statutory regulation RAG subsystem indexes municipal gazettes, development plans, and building bye-laws using a hybrid property graph

and vector database, providing legally cited answers. The document intelligence pipeline extracts structured tables, clauses, and vector coordinates from scanned municipal PDFs and survey maps. The enterprise reporting generator compiles comprehensive feasibility prospectuses, investment memoranda, and vector CAD/BIM exports.

The collaboration subsystem synchronizes multi-user editing sessions in real time via WebSockets and Conflict-Free Replicated Data Types (CRDTs), preventing data conflicts during concurrent parcel modifications. The audit and versioning engine logs an immutable record of every computational run, capturing the exact code versions, rule sets, geometric inputs, and user overrides to ensure legal auditability and reproducibility. Telemetry pipelines capture performance traces, geometric failures, and API latencies. Finally, the administrative rule portal provides municipal architects with curation tools to draft, validate, and publish version-controlled statutory rule packages.

System Architecture Evaluation and Topology

A platform combining real-time canvas drafting (60Hz interaction loops), non-linear geometric computations (straight skeleton offsetting, Minkowski dilations), and combinatorial optimizations requires careful architectural decoupling. Five candidate architectural models were evaluated to determine the appropriate system topology.

Architecture Comparison

Architecture A (Monolith) houses the web server, drawing controllers, geometry libraries, regulatory logic, and financial calculations within a single runtime (such as a Node.js or Python process). While providing rapid initial development speed and zero network serialization overhead, heavy CPU operations—such as calculating straight skeleton offsets or running genetic layout loops—block the main thread, causing API latencies and risking memory exhaustion across all active tenants.

Architecture B (Modular Monolith) partitions application logic into strictly bounded domains (Identity, Projects, Geometry, Regulations, Finance) communicating via in-memory interfaces within a unified codebase. This pattern provides architectural isolation, simplified local testing, and atomic ACID transactions within a single relational database. However, scaling requires replicating the entire application footprint, and resource contention between I/O-bound web traffic and CPU-bound spatial mathematics persists.

Architecture C (Microservices) isolates every domain into independently deployed services communicating via synchronous HTTP/REST and gRPC over an API Gateway. While offering independent horizontal scaling and service-level language optimization (such as Go for networking and C++ for geometry), it introduces substantial operational complexity, distributed transaction management challenges, network serialization latencies for large GeoJSON geometry payloads, and brittle inter-service dependencies.

Architecture D (Event-Driven Computation Platform) decouples services around a distributed event log (such as Apache Kafka). State mutations are published as immutable domain events, with downstream computation engines consuming streams asynchronously. While well-suited for eventual consistency and long-running batch analytics, this pattern adds unnecessary architectural overhead to interactive user workflows, complicating immediate auto-save feedback and real-time canvas interactions.

Architecture E (Hybrid Application with Asynchronous Computation Workers) combines an

I/O-optimized Application Core with an asynchronous pool of horizontally scalable Computation Workers managed via Redis Streams and Celery. The Application Core handles client authentication, project metadata, regulatory evaluations, and financial calculations, while compute-heavy workloads (straight skeleton generation, generative layout synthesis, multi-objective optimization, and PDF generation) are offloaded to isolated background worker containers. This architecture provides responsive user interactions alongside dedicated, scalable compute resources for heavy mathematical processing.

Architectural Evaluation Matrix

The evaluation across architectural patterns is detailed below.

Architectural Dimension	Architecture A: Monolith	Architecture B: Modular Monolith	Architecture C: Microservices	Architecture D: Event-Driven	Architecture E: Hybrid App + Workers (Selected)
Development Speed (MVP)	Very High	High	Low	Low	High
System Complexity	Very Low	Low	Very High	High	Moderate
Horizontal Scalability	Low	Moderate	High	Very High	High (Targeted worker scaling)
Debugging & Traceability	High	High	Low	Moderate	High
Deployment Simplicity	Very High	High	Low	Moderate	High
Data Consistency Model	ACID (Single DB)	ACID (Single DB)	Eventual Consistency	Eventual Consistency	Strong for core; Eventual for jobs
Heavy Computational Handling	Poor (Blocks I/O)	Poor (Blocks runtime)	Good	High	Excellent (Worker isolation)
Optimal Engineering Team Size	2–5 Engineers	4–8 Engineers	20+ Engineers	15+ Engineers	6–12 Engineers
Cloud Cost Profile	Lowest	Low	High	High	Optimized (Compute on demand)

Phased System Evolution

To ensure rapid delivery without incurring premature technical debt, the platform evolves through three structured deployment phases:

- *MVP Phase (Days 1–90)*: The platform deploys as an Architecture B/E hybrid. An application core handles authentication, project metadata, and light spatial interactions, coupled via a local Redis queue to a standalone Python/C++ geometry worker container running CGAL and GEOS operations.
- *Growth Phase (Months 4–12)*: The system transitions to a fully decoupled Architecture E topology. The application core scales behind a load balancer, routing computational tasks

to dedicated worker pools: a Python Geometry Worker pool, a C++ Layout Optimization pool, and an asynchronous Document Intelligence OCR pool. All state persists to a managed PostgreSQL/PostGIS database cluster.

- *Enterprise Phase (Year 2+)*: The platform implements enterprise multi-tenancy, with tenant-isolated database schemas managed via PostgreSQL Row-Level Security, a dedicated gRPC microservice for institutional financial modeling, and distributed GPU worker clusters running deep learning layout synthesis and real-time daylighting simulations.

Production System Architecture Topology

The production architecture is structured into six functional tiers that isolate user interactions from compute-intensive operations:

1. *Presentation Tier*: Modern Web Browsers running React 19, TypeScript, and a unified MapLibre/Three.js WebGL canvas context. Manages user pointer events, dynamic toolbars, and HUD metric readouts.
2. *Ingress & Security Perimeter*: Cloudflare Enterprise CDN providing DDoS mitigation, WAF rules, and SSL/TLS 1.3 termination. Routes to an internal Kong API Gateway enforcing rate limiting, CORS headers, and OAuth2/JWT cryptographic validation.
3. *Application Core*: Horizontally scaled Node.js (NestJS) or Go instances managing tenant metadata, user permissions, project versioning, and client WebSocket connections.
4. *High-Speed Cache & Message Broker*: Redis 7.2 cluster providing in-memory session caching, spatial boundary caching, and job queuing via Redis Streams and Celery.
5. *Computation Worker Tier*: Containerized Python and C++ background workers consuming jobs from queues. Integrates CGAL 6.0, GEOS 3.12, Shapely 2.0, and Google OR-Tools to execute straight skeleton offsetting, boolean polygon clipping, generative unit packing, and multi-objective optimization without blocking the application core.
6. *Persistence Tier*: High-availability PostgreSQL 16 cluster with PostGIS 3.4 spatial extensions and TimescaleDB for audit telemetry. Utilizes AWS S3 for binary storage of CAD drawings, IFC files, and compiled PDF reports.

Technology Stack Evaluation and Trade-Off Analysis

Selecting the enterprise software stack requires balancing computational performance, numerical accuracy, spatial library maturity, licensing terms, and integration complexity.

Technology Selection and Evaluation Matrix

The technical layers are evaluated across established options to substantiate architectural choices.

Layer / Technology	Candidates Evaluated	Selected Technology	Technical Trade-Offs and Justification
Frontend Framework	React 19, Next.js 14, Vite + React, SvelteKit	Vite + React 19 + TypeScript	Next.js adds unnecessary SSR complexity for authenticated, client-side WebGL

Layer / Technology	Candidates Evaluated	Selected Technology	Technical Trade-Offs and Justification
			workspaces. Vite provides fast HMR and lightweight bundle sizes. TypeScript guarantees compile-time type safety across complex GeoJSON coordinate structures.
Client State & Cache	Redux Toolkit, Zustand, TanStack Query, MobX	TanStack Query v5 + Zustand	TanStack Query isolates server-state caching, optimistic updates, and background refetching. Zustand handles ephemeral UI state (active tool modes, modal flags) with minimal boilerplate and zero unnecessary re-renders.
Client GIS Rendering	Mapbox GL JS v3, MapLibre GL JS, CesiumJS, OpenLayers	MapLibre GL JS v4 / Mapbox GL JS	Mapbox GL JS offers mature 3D terrain and satellite basemaps but carries proprietary pricing. MapLibre provides a fully open-source (BSD-3), high-performance WebGL/WebGPU fork with zero vendor lock-in, supporting custom Three.js layers.
Backend Core	Node.js (NestJS), Python (FastAPI), Go, Django	FastAPI (Python 3.11) + NestJS (Node.js)	Node.js/NestJS excels at I/O-bound CRUD, WebSockets, and enterprise architectures. FastAPI natively interfaces with C-based scientific and geospatial libraries (NumPy, Shapely, PyProj) with high-performance asynchronous

Layer / Technology	Candidates Evaluated	Selected Technology	Technical Trade-Offs and Justification
			execution.
Spatial Database	PostgreSQL/PostGIS, MongoDB, Spatialite, CockroachDB	PostgreSQL 16 + PostGIS 3.4	PostGIS is the global standard for enterprise spatial databases, offering OGC-compliant spatial predicates, GiST spatial indexing, robust projection engines (ST_Transform), and native topological validation.
Computational Geometry	CGAL, GEOS, JTS, Clipper2, Turf.js, Shapely	CGAL 6.0 (C++) + GEOS / Shapely 2.0 (Python)	Turf.js is limited to simple client-side geodesic math. GEOS/Shapely provides robust 2D planar booleans. CGAL is required for mathematically exact straight skeleton inward offsets, mitered polygon buffering, and Delaunay triangulations without topological failures.
3D Visualization	Three.js, Babylon.js, CesiumJS, Deck.gl	Three.js (r162+) inside Mapbox/MapLibre	CesiumJS is optimized for planetary-scale GIS but heavy for architectural parcels. Babylon.js is game-focused. Three.js provides lightweight procedural mesh generation, CSG booleans, and integration directly into Mapbox WebGL contexts.
Optimization Solvers	Google OR-Tools, SciPy, Pyomo, Optuna, Platypus	Google OR-Tools + Platypus / DEAP (NSGA-II)	OR-Tools provides high-performance Mixed-Integer Linear Programming (MILP) for floor plate unit packing.

Layer / Technology	Candidates Evaluated	Selected Technology	Technical Trade-Offs and Justification
			Platypus/DEAP implements NSGA-II for multi-objective evolutionary optimization along the Pareto frontier.
AI / RAG Infrastructure	LlamaIndex, LangChain, Neo4j, Qdrant, OpenAI / Anthropic	<i>Qdrant (Vector) + Neo4j (Graph) + Claude 3.5 Sonnet</i>	Pure vector search fails on hierarchical legal structures. A GraphRAG architecture combining Neo4j (statutory hierarchy) and Qdrant (semantic vector chunks) enables legally cited, explainable regulatory retrieval.
Message Broker & Jobs	RabbitMQ, Apache Kafka, Redis Streams, Celery	Redis 7.2 (Streams) + Celery / ARQ	Kafka adds high operational overhead for early-to-mid stage scale. Redis Streams paired with Celery provides lightweight, low-latency task distribution for asynchronous geometry and feasibility jobs.

Canonical Domain Data Model

The domain architecture models all physical, legal, and financial entities as strongly typed, versioned domain structures. Relationships flow downward from multi-tenant enterprise organizations to individual cadastral parcels, regulatory compilations, building configurations, and financial pro-formas.

Entity Relational Structure

The entity relationships are structured as follows:

- An Organization owns multiple Users and Projects.
- A Project aggregates one or more Sites and contains multiple versioned Development Scenarios.
- A Site contains one or more Cadastral Parcels and abuts one or more Road Segments.
- A Parcel maintains an outer Boundary and contains multiple Site Constraints (easements, utility reservations, tree protection markers).
- A Development Scenario references a specific Regulation Version and compiles the

- spatial deductions into a 3D Buildable Envelope.
- The Scenario synthesizes one or more Buildings, a Financial Model, and an immutable Audit Trail.
- Each Building decomposes vertically into Floors, which house residential Units, circulation Corridors, structural Cores (elevators, stairs, MEP shafts), and Parking bays.
- Each Financial Model compiles dynamic Cost Items, Revenue Items, and periodic Cash Flow schedules to derive project yields (IRR, NPV, RLV).

Master Entity Attributes and Specifications

The master specifications defining the system's core domain entities are detailed below.

Entity Name	Primary Purpose and Lifecycle	Key Attributes and Types	Foreign Relationships & Ownership	Versioning & Validation Rules
User	Authentication identity and access credentials.	id (UUID), email (CI-String), role (Enum), password_hash (String).	Belongs to an Organization.	Immutable ID; validates RFC 5322 email formatting.
Organization	Multi-tenant administrative isolation container.	id (UUID), name (String), plan_tier (Enum), created_at (Timestamp).	Owns Users, Projects, and Rulesets.	Soft-delete retention policy; unique enterprise slug.
Project	Encapsulates the development brief and parcel portfolio.	id (UUID), title (String), tenure_type (Enum), version (Int).	Owned by Organization; references active Scenario.	Monotonically incrementing version; validates title length.
Site	Aggregates contiguous cadastral parcels into a single site.	id (UUID), name (String), gross_area_sqm (Numeric), centroid (Point).	Belongs to Project; owns Parcels and Roads.	Gross area must equal the sum of constituent parcels.
Parcel	Individual legal revenue survey property.	id (UUID), survey_number (String), boundary_wgs84 (Geom), srid (Int).	Belongs to Site; owns Boundary and Constraints.	OGC Simple Feature validation (ST_IsValid asserted).
Boundary	Topological outer perimeter of a land parcel.	id (UUID), geom_wgs84 (Polygon), geom_projected (Polygon), is_valid (Bool).	Owned by Parcel.	Counter-clockwise winding; ring closure asserted.
RoadSegment	Abutting transit corridors	id (UUID), road_name	Abuts Site and Parcels.	Width > 0; requires at least one

Entity Name	Primary Purpose and Lifecycle	Key Attributes and Types	Foreign Relationships & Ownership	Versioning & Validation Rules
	establishing access widths.	(String), width_m (Numeric), centerline (LineString).		primary access edge.
Constraint	Spatial exclusions: easements, tree buffers, reservations.	id (UUID), type (Enum), buffer_m (Numeric), geom_projected (Geometry).	Owned by Parcel.	Must reside completely within parcel boundary.
RegulationVersion	Machine-readable municipal planning ruleset.	id (UUID), jurisdiction (String), code (String), rules_dsl (JSONB).	Curated by Admin; evaluated by Scenarios.	Semantically versioned; immutable once published.
RuleResult	Explainable audit record for a statutory evaluation.	id (UUID), rule_id (String), is_compliant (Bool), citation (Text).	Linked to Scenario and RegulationVersion.	Immutable execution record; captures gazette clause.
BuildableEnvelope	Net 3D polyhedral envelope available for construction.	id (UUID), footprint_2d (Polygon), max_height_m (Numeric), volume_3d (Geom).	Derived from Parcel, Constraints, and Rules.	Footprint area must be ≤ net parcel area.
DevelopmentScenario	Specific design branch evaluated on a parcel.	id (UUID), name (String), target_fsi (Numeric), status (Enum).	Owned by Project; owns Buildings and Pro-Forma.	Branching DAG lifecycle (Draft, Running, Complete).
Building	Discrete structural tower or podium block.	id (UUID), name (String), footprint (Polygon), height_m (Numeric).	Owned by Scenario; owns Floors.	Footprint must reside within BuildableEnvelope.
Floor	Horizontal architectural level within a building.	id (UUID), floor_number (Int), floor_type (Enum), bua_sqm (Numeric).	Owned by Building; owns Units and Cores.	Level elevation must match floor-to-floor heights.
Unit	Individual residential or commercial apartment.	id (UUID), unit_number (String), typology (Enum),	Owned by Floor.	Carpet area conforms to statutory NBC 2016 norms.

Entity Name	Primary Purpose and Lifecycle	Key Attributes and Types	Foreign Relationships & Ownership	Versioning & Validation Rules
		carpet_sqm (Numeric).		
StructuralGrid	Parametric column layout and structural transfer spans.	id (UUID), grid_x_m (Numeric), grid_y_m (Numeric), transfer_flag (Bool).	Linked to Building and Floors.	Regular spans (6.0m–7.5m); identifies transfers.
Parking	Vehicular stalls, ramps, and circulation aisles.	id (UUID), car_stalls (Int), ev_bays (Int), aisle_geom (MultiPolygon).	Owned by Floor.	Stall counts must meet municipal statutory quotas.
FinancialModel	Institutional DCF underwriting pro-forma.	id (UUID), gdv (Numeric), tdc (Numeric), irr (Numeric), rlv (Numeric).	Owned by DevelopmentScenario.	Deterministic mathematical execution; zero AI drift.
AuditEvent	Cryptographic audit log of engine computations.	id (UUID), engine_versions (JSONB), input_hash (Char), timestamp (Time).	Global immutable log.	Append-only partition; cryptographically signed.

Spatial Database Design and PostGIS Implementation

Storing spatial coordinates as raw stringified GeoJSON strings inside generic document columns degrades performance by eliminating database-level spatial indexing (R-Tree / GiST), preventing spatial integrity checks, and shifting coordinate transformations to the client.

The platform implements a hybrid spatial persistence model:

- Primary spatial entities persist dual geometries in PostgreSQL 16 with PostGIS 3.4 extensions: an ellipsoidal geometry (EPSG:4326) for global API serialization, and a local projected metric geometry (EPSG:UTM Zone) for spatial mathematics.
- Spatial indexes (GiST) are created on all projected geometries, enabling efficient bounding-box queries (ST_Intersects, ST_Contains) across large cadastral layers.
- Generated floor plates, unit layouts, and temporary straight skeleton bisector trees are cached in Redis and persisted to relational child tables only when a design run is finalized.

Production PostGIS DDL Schema

-- Extensions

CREATE EXTENSION IF NOT EXISTS "uuid-oss";

CREATE EXTENSION IF NOT EXISTS "postgis";

```
CREATE EXTENSION IF NOT EXISTS "btree_gist";
```

```
-- Custom Enums
```

```
CREATE TYPE land_tenure_type AS ENUM (  
    'PRIVATE_FREEHOLD',  
    'COLLECTOR_CLASS_2',  
    'COLLECTOR_LEASEHOLD',  
    'GOVERNMENT_LEASEHOLD',  
    'MHADA_REDEVELOPMENT',  
    'MUNICIPAL_ALLOTMENT'  
);
```

```
CREATE TYPE constraint_severity_type AS ENUM (  
    'HARD_ABSOLUTE_EXCLUSION',  
    'CONDITIONAL_NOC_REQUIRED',  
    'HEIGHT_RESTRICTION_SURFACE',  
    'SETBACK_OFFSET_MODIFIER'  
);
```

```
CREATE TYPE unit_typology_type AS ENUM (  
    'STUDIO', '1BHK', '2BHK', '3BHK', '4BHK', 'RETAIL', 'OFFICE'  
);
```

```
-- Parcels Table
```

```
CREATE TABLE parcels (  
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),  
    organization_id UUID NOT NULL,  
    project_id UUID NOT NULL,  
    cadastral_survey_number VARCHAR(128) NOT NULL,  
    tenure_type land_tenure_type NOT NULL DEFAULT 'PRIVATE_FREEHOLD',  
    srid_projected INT NOT NULL,  
    boundary_wgs84 GEOMETRY(MultiPolygon, 4326) NOT NULL,  
    boundary_projected GEOMETRY(MultiPolygon) NOT NULL,  
    gross_area_sqm NUMERIC(14, 4) GENERATED ALWAYS AS  
(ST_Area(boundary_projected)) STORED,  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
    version INT NOT NULL DEFAULT 1,  
    is_deleted BOOLEAN NOT NULL DEFAULT FALSE,  
    CONSTRAINT chk_boundary_wgs84_valid CHECK (ST_IsValid(boundary_wgs84)),  
    CONSTRAINT chk_boundary_projected_valid CHECK (ST_IsValid(boundary_projected))  
);
```

```
CREATE INDEX idx_parcels_boundary_wgs84 ON parcels USING GIST (boundary_wgs84);  
CREATE INDEX idx_parcels_boundary_projected ON parcels USING GIST  
(boundary_projected);  
CREATE INDEX idx_parcels_org_proj ON parcels (organization_id, project_id) WHERE NOT  
is_deleted;
```

-- Parcel Roads Table

```
CREATE TABLE parcel_roads (  
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),  
  parcel_id UUID NOT NULL REFERENCES parcels(id) ON DELETE CASCADE,  
  road_name VARCHAR(256),  
  existing_width_m NUMERIC(6, 2) NOT NULL CHECK (existing_width_m > 0),  
  proposed_width_m NUMERIC(6, 2) CHECK (proposed_width_m >= existing_width_m),  
  road_centerline GEOMETRY(LineString, 4326),  
  road_boundary_edge GEOMETRY(LineString, 4326) NOT NULL,  
  is_primary_access BOOLEAN NOT NULL DEFAULT TRUE,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);
```

```
CREATE INDEX idx_parcel_roads_edge ON parcel_roads USING GIST  
(road_boundary_edge);
```

-- Parcel Constraints Table

```
CREATE TABLE parcel_constraints (  
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),  
  parcel_id UUID NOT NULL REFERENCES parcels(id) ON DELETE CASCADE,  
  constraint_name VARCHAR(128) NOT NULL,  
  constraint_category VARCHAR(64) NOT NULL,  
  severity constraint_severity_type NOT NULL,  
  buffer_distance_m NUMERIC(6, 2) NOT NULL DEFAULT 0.0,  
  vertical_clearance_limit_m NUMERIC(8, 2),  
  footprint_wgs84 GEOMETRY(Geometry, 4326) NOT NULL,  
  footprint_projected GEOMETRY(Geometry) NOT NULL,  
  buffer_geometry_projected GEOMETRY(Geometry) NOT NULL,  
  statutory_reference VARCHAR(256),  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);
```

```
CREATE INDEX idx_parcel_constraints_geom ON parcel_constraints USING GIST  
(buffer_geometry_projected);
```

-- Parcel Trees Table

```
CREATE TABLE parcel_trees (  
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),  
  parcel_id UUID NOT NULL REFERENCES parcels(id) ON DELETE CASCADE,  
  tag_number VARCHAR(64),  
  species VARCHAR(128),  
  trunk_girth_m NUMERIC(4, 2),  
  canopy_radius_m NUMERIC(4, 2) NOT NULL DEFAULT 3.0,  
  mandatory_root_buffer_m NUMERIC(4, 2) NOT NULL DEFAULT 2.5,  
  location_wgs84 GEOMETRY(Point, 4326) NOT NULL,  
  location_projected GEOMETRY(Point) NOT NULL,  
  is_felling_prohibited BOOLEAN NOT NULL DEFAULT TRUE,
```

```

    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_parcel_trees_wgs84 ON parcel_trees USING GIST (location_wgs84);

-- Financial Models Table
CREATE TABLE financial_models (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    scenario_id UUID NOT NULL UNIQUE,
    gdv_total NUMERIC(16, 2) NOT NULL,
    tdc_total NUMERIC(16, 2) NOT NULL,
    gross_profit NUMERIC(16, 2) GENERATED ALWAYS AS (gdv_total - tdc_total) STORED,
    return_on_cost_pct NUMERIC(6, 4) GENERATED ALWAYS AS (
        CASE WHEN tdc_total > 0 THEN ((gdv_total - tdc_total) / tdc_total) * 100 ELSE 0 END
    ) STORED,
    irr_annualized NUMERIC(6, 4) NOT NULL,
    npv NUMERIC(16, 2) NOT NULL,
    residual_land_value NUMERIC(16, 2) NOT NULL,
    cash_flow_schedule JSONB NOT NULL DEFAULT '[]'::jsonb,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_financial_models_scenario ON financial_models(scenario_id);

-- Immutable Calculation Audit Trail Table (Partitioned by Year)
CREATE TABLE calculation_audit_logs (
    id UUID NOT NULL DEFAULT uuid_generate_v4(),
    organization_id UUID NOT NULL,
    project_id UUID NOT NULL,
    scenario_id UUID NOT NULL,
    engine_name VARCHAR(64) NOT NULL,
    engine_version VARCHAR(32) NOT NULL,
    rule_version VARCHAR(32) NOT NULL,
    input_parameters_hash CHAR(64) NOT NULL,
    output_metrics_hash CHAR(64) NOT NULL,
    raw_payload_json JSONB NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    PRIMARY KEY (id, created_at)
) PARTITION BY RANGE (created_at);

CREATE TABLE audit_logs_2025 PARTITION OF calculation_audit_logs
    FOR VALUES FROM ('2025-01-01') TO ('2026-01-01');

CREATE INDEX idx_audit_logs_lookup ON calculation_audit_logs (organization_id, project_id,
    created_at DESC);

```

Coordinate Reference System and GeoJSON Data Contract

A critical failure mode in early spatial software prototypes is executing statutory setbacks, parcel areas, and buffer distances in non-conformal projections. Calculating areas directly in Web Mercator (EPSG:3857) introduces severe latitude-dependent geometric distortion: $k = \sec(\phi) = \frac{1}{\cos(\phi)}$. Because planar area scales with the square of the linear scale factor: $k_{\text{area}} = \sec^2(\phi)$. At a latitude of 19.115° N (representative of Mumbai, India), computing parcel areas in Web Mercator artificially inflates the square footage: $k_{\text{area}} = \sec^2(19.115^\circ) = \left(\frac{1}{\cos(19.115^\circ)}\right)^2 \approx \left(\frac{1}{0.94488}\right)^2 \approx 1.1200$ (approx +12.00% area error). Underwriting multi-million-dollar land transactions or computing statutory setbacks using Web Mercator produces legally invalid results. The platform enforces a four-tier Coordinate Reference System strategy:

1. *Map Rendering Tier*: MapLibre GL JS operates on WGS84 geographic coordinates (EPSG:4326), projecting them to Web Mercator (EPSG:3857) on the GPU solely for raster basemap display.
2. *Automated Conformal Projection*: When a parcel boundary is received, the backend computes its geographic centroid and dynamically derives the corresponding Universal Transverse Mercator (UTM) metric projection: $\text{UTM Zone} = \left\lfloor \frac{\lambda}{6} + 180 \right\rfloor + 1$. $\text{EPSG Code} = \begin{cases} 32600 + \text{UTM Zone} & \text{for Northern Hemisphere } (\phi \geq 0) \\ 32700 + \text{UTM Zone} & \text{for Southern Hemisphere } (\phi < 0) \end{cases}$. For Mumbai (72.85° E), the system assigns UTM Zone 43N (EPSG:32643).
3. *Computational Geometry Isolation*: All straight skeleton offsets, Minkowski buffers, road widening subtractions, and usable area integrations occur strictly in metric Cartesian space.
4. *Serialization Contract*: Geometries are reprojected back to WGS84 (EPSG:4326) with six decimal places of precision ($\sim 0.11 \text{ m}$ resolution) only when serializing payloads for REST/WebSocket delivery.

Canonical GeoJSON Feature Schema

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "ParcelFeatureCollectionContract",
  "type": "object",
  "required": ["type", "features", "crs_metadata"],
  "properties": {
    "type": { "type": "string", "enum": ["FeatureCollection"] },
    "crs_metadata": {
      "type": "object",
      "required": ["canonical_srid", "projected_srid", "utm_zone"],
      "properties": {
        "canonical_srid": { "type": "integer", "enum": [4326] },
        "projected_srid": { "type": "integer" },

```

```

    "utm_zone": { "type": "string" }
  },
  "features": {
    "type": "array",
    "items": {
      "type": "object",
      "required": ["type", "id", "geometry", "properties"],
      "properties": {
        "type": { "type": "string", "enum": ["Feature"] },
        "id": { "type": "string", "format": "uuid" },
        "geometry": {
          "type": "object",
          "required": ["type", "coordinates"],
          "properties": {
            "type": {
              "type": "string",
              "enum": ["Polygon", "MultiPolygon", "Point", "LineString"]
            },
            "coordinates": { "type": "array" }
          }
        },
        "properties": {
          "type": "object",
          "required": ["entity_type", "version", "updated_at"],
          "properties": {
            "entity_type": {
              "type": "string",
              "enum": [
                "SITE_BOUNDARY",
                "ROAD_WIDENING_DEDUCTION",
                "EASEMENT_DEDUCTION",
                "PRESERVED_TREE",
                "BUILDABLE_ENVELOPE",
                "BUILDING_FOOTPRINT"
              ]
            },
            "version": { "type": "integer" },
            "updated_at": { "type": "string", "format": "date-time" },
            "accuracy_confidence": { "type": "number", "minimum": 0, "maximum": 1 }
          }
        }
      }
    }
  }
}

```

Computational Geometry Engine Specification

The Computational Geometry Engine provides exact mathematical execution for all spatial operations. The platform couples the Computational Geometry Algorithms Library (CGAL 6.0) for straight skeleton inward offsetting with GEOS and Clipper2 for planar booleans and decomposition.

Master Computational Geometry Operations Matrix

The algorithmic and mathematical specifications for the primary spatial operations are detailed below.

#	Operation & Mathematical Definition	Algorithmic Strategy & Complexity	Edge Cases & Validation Rules	Engine & Library Implementation
1	Planar Polygon Area $A = \frac{1}{2} \left \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i) \right $	Shoelace Formula / Green's Theorem discrete summation. $\mathcal{O}(n)$.	Zero-width self-tangent corridors; collinear segments. Must assert simplicity before calculation.	GEOS / Shapely (shapely.area) / PostGIS (ST_Area).
2	Spherical Geodesic Area $E = \sum_{j=1}^k \alpha_j - (k-2)\pi$, $A = R^2 E$	Spherical excess integration over the authalic sphere. $\mathcal{O}(n)$.	Antimeridian crossing; pole encapsulation; self-intersecting loops.	GeographicLib / PostGIS (ST_Area(geog, true)).
3	Perimeter Summation $P = \sum_{i=0}^{n-1} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}$	Euclidean distance summation over metric vertices. $\mathcal{O}(n)$.	Consecutive duplicate vertices; micro-spikes where $\Delta d < 10^{-6} \text{m}$.	GEOS / Shapely (shapely.length).
4	Centroid (Center of Mass) $C_x = \frac{1}{6A} \sum (x_i + x_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$	Second moment of area polygon integration. $\mathcal{O}(n)$.	Centroid of concave/horseshoe polygons falls outside the polygon interior.	GEOS (ST_Centroid) / CGAL 2D Polygon.
5	Point of Inaccessibility $\max_p \min_{e \in \text{partial } P} \text{dist}(p, e)$	Quadtree space-partitioning scan (Polylabel algorithm). $\mathcal{O}(n \log n)$.	Highly elongated or branching geometries with multiple local optima.	Mapbox Polylabel / Custom C++ Quadtree.
6	Minimum Rotated BBox $\min_{\theta}$	Rotating Calipers algorithm	Collinear degenerate hulls;	GEOS (ST_OrientedEnve

#	Operation & Mathematical Definition	Algorithmic Strategy & Complexity	Edge Cases & Validation Rules	Engine & Library Implementation
	$(W_{\theta} \times H_{\theta}) \text{ s.t. } P \subseteq R_{\theta}$	traversing convex hull edges. $\mathcal{O}(n \log n)$.	square symmetry (90° orientation ambiguity).	lope) / Shapely.
7	Polygon Orientation $\text{sgn}(\sum_{i=1}^n (x_{i+1} - x_i)(y_{i+1} + y_i))$	Trapezoidal signed area integration. $\mathcal{O}(n)$.	Flat collinear loops where signed area equals zero.	CGAL / Shapely (shapely.ops.orient).
8	Self-Intersection Check $\exists s_i, s_j \text{ s.t. } s_i \cap s_j \neq \emptyset$	Bentley-Ottmann sweep-line segment intersection algorithm. $\mathcal{O}((n+k) \log n)$.	Touching vertices at ring start/end; collinear overlapping edges.	GEOS (ST_IsSimple, ST_IsValid).
9	Topological Validation OGC Simple Feature Rules enforcement.	GEOS topological consistency validator. $\mathcal{O}(n \log n)$.	Inverted interior holes; nested self-touching rings; spike slivers.	PostGIS (ST_IsValidReason, ST_MakeValid).
10	Boolean Union $P_A \cup P_B = \{p \mid p \in P_A \vee p \in P_B\}$	Martinez-Rueda-Feito boolean union sweep-line algorithm. $\mathcal{O}(n \log n)$.	Disjoint non-overlapping polygons return a MultiPolygon.	Clipper2 / GEOS (ST_Union).
11	Boolean Intersection $P_A \cap P_B = \{p \mid p \in P_A \wedge p \in P_B\}$	Weiler-Atherton / Martinez-Rueda graph traversal. $\mathcal{O}(n \log n)$.	Tangent coplanar edges yielding degenerate linear/point intersections.	Clipper2 / GEOS (ST_Intersection).
12	Boolean Difference $P_A \setminus P_B = \{p \mid p \in P_A \wedge p \notin P_B\}$	Planar topological graph clipping. $\mathcal{O}(n \log n)$.	Subtractions that bisect a polygon into two or more disjoint islands.	Clipper2 / GEOS (ST_Difference).
13	Straight Skeleton Offsets Wavefront grassfire inward propagation.	CGAL Straight Skeleton 2D construction via edge/split events. $\mathcal{O}(n \log n)$.	Non-uniform edge setback weights; extreme acute angles producing infinite bisectors.	CGAL 6.0 (CGAL::Straight_skeleton_2).

#	Operation & Mathematical Definition	Algorithmic Strategy & Complexity	Edge Cases & Validation Rules	Engine & Library Implementation
14	Minkowski Sum / Dilation $A \oplus B = \{a + b \mid a \in A, b \in B\}$	Convolution of polygon contours with circular polygonal disk. $\mathcal{O}(n \cdot m)$.	Merging of adjacent disjoint holes when buffer distance is large.	CGAL 2D Polygon Offset / Clipper2.
15	Setback Generation $P \setminus \bigcup_{i=1}^n \text{HalfPlane}(e_i, d_i)$	Weighted straight skeleton inward offsetting per edge class. $\mathcal{O}(n \log n)$.	Parcels where setbacks exceed plot width, collapsing envelope to null.	CGAL 6.0 Weighted Offset Builder.
16	Road Widening Deduction $P_{\text{site}} \setminus \text{Buffer}(\text{Centerline}, W_{\text{new}}/2)$	Boolean difference between parcel and road widening polygon corridor. $\mathcal{O}(n \log n)$.	Curved road centerlines intersecting non-frontage side boundaries.	GEOS / PostGIS (ST_Difference).
17	Tree Root Preservation Buffer $B(p_{\text{tree}}, r_{\text{root}}) = \{x \mid \text{Vert}(x) - p_{\text{tree}} \leq r_{\text{root}}\}$	Circular Minkowski disk generation. $\mathcal{O}(k)$.	Tree buffers intersecting external site boundaries; overlapping trees.	Shapely (shapely.buffer(cap_style=1)).
18	Exclusion Zone Clipping $P_{\text{buildable}} = P_{\text{site}} \setminus \bigcup_j \text{Exclusions}_j$	Sequential multi-layer boolean difference pipeline. $\mathcal{O}(m \cdot n \log n)$.	Multi-ring outputs; disconnected slivers below threshold.	GEOS / Clipper2.
19	Buildable Envelope Extraction $\text{Clean}(P_{\text{te xt{envelope}}}) = \{P_k \mid A(P_k) \geq A_{\min}\}$	Topological difference filtered by minimum usable area threshold. $\mathcal{O}(k)$.	Complete envelope collapse; fragmented disjoint sub-footprints.	Custom Geometry Pipeline.
20	Minimum Usable Width Scan $\min_{p \in \text{MedialAxis}}$	Medial axis Voronoi transformation scanning inscribed	Narrow bottlenecks connecting two large, viable	CGAL Voronoi / Boost.Polygon.

#	Operation & Mathematical Definition	Algorithmic Strategy & Complexity	Edge Cases & Validation Rules	Engine & Library Implementation
	$2 \cdot r_{\text{inscribed}}(p)$	circle radii.	footprint zones.	
21	Medial Axis / Skeleton $\text{MA}(P) = \{x \in P \mid \exists c_1, c_2 \in \partial P, \text{Vert}(x-c_1) = \text{Vert}(x-c_2)\}$	Voronoi diagram over densified boundary vertices. $\mathcal{O}(n \log n)$.	Boundary noise generating spurious skeletal branches.	Boost.Polygon Voronoi Diagram.
22	Convex Hull Generation Minimal convex set enclosing polygon vertices.	Monotone Chain (Andrew's variant of Graham Scan). $\mathcal{O}(n \log n)$.	Collinear boundary points; degenerate flat line polygons.	GEOS (ST_ConvexHull) / CGAL.
23	Optimal Inscribed Rectangle $\max(w \cdot h) \text{ s.t. } R(w, h, \theta) \subseteq P_{\text{envelope}}$	Discrete raster grid branch-and-bound optimization. $\mathcal{O}(n^2)$.	Highly irregular, non-convex shapes yielding multiple local optima.	Custom CGAL Branch-and-Bound solver.
24	Convex Decomposition $P = \bigcup_{i=1}^m C_i \text{ s.t. } C_i \text{ is convex}$	Hertel-Mehlhorn triangulation reduction algorithm. $\mathcal{O}(n \log n)$.	Over-partitioning generating narrow, non-constructible triangles.	CGAL 2D Polygon Partitioning.
25	Vehicle Swept-Path Sweep Continuous envelope along Ackerman steering curve.	Reeds-Shepp vehicle trajectory curve generation with polygonal dilation.	Tight 90° corners clipping curbs; reverse maneuvers.	Custom Python Kinematic Model.

Straight Skeleton Offsetting Engine

Municipal planning codes mandate non-uniform perimeter setbacks tied to road frontages, rear

boundaries, and lateral property lines. Under Mumbai DCPR 2034 Table 12, a parcel fronting an 18.0m road requires a 4.5m Front Setback, a 3.0m Rear Setback, and 3.0m Side Setbacks. Conventional buffering algorithms—such as Minkowski sums implemented in GEOS or Turf.js—round exterior convex corners, distorting the buildable envelope and miscalculating potential floor plates.

The platform utilizes CGAL's weighted straight skeleton engine (CGAL::Straight_skeleton_2). The straight skeleton is defined by an inward grassfire wavefront propagation spawned from every boundary edge e_i at time $t = 0$. Each edge advances inward parallel to its initial orientation at a speed proportional to its statutory setback weight w_i . Moving vertices trace angular bisectors between adjacent edges.

The advancing wavefront undergoes topological transitions driven by two core geometric events:

- *Edge Event*: An advancing edge contracts to zero length as its incident vertices collide, causing the edge to vanish and reducing the polygon's vertex count.
- *Split Event*: An advancing reflex vertex (internal angle $> \pi$) collides with an advancing non-adjacent edge, splitting the polygon into two independent sub-polygons that continue contracting independently.

The straight skeleton partitions the interior of the polygon into monotone planar faces, with each face associated with a specific boundary edge. By evaluating the angular bisector rays at statutory setback distances ($t_i = \text{Setback}_i$), the engine extracts inward offset polygons with sharp, mitered corners conforming to architectural construction practices.

Declarative Regulation Engine Specification

Regulations are modeled as declarative, version-controlled software artifacts rather than embedded procedural logic. When municipal authorities publish gazette notifications amending building bye-laws, developers can evaluate parcels against both current and proposed regulations without modifying application code.

Multi-Jurisdictional Statutory Reference Matrix

The engine encodes development control rules across major municipal jurisdictions:

State / Region	Planning Authority	Governing Statutory Code & Clauses	Base FSI	Maximum Permissible FSI	Road Width Thresholds	Statutory Setback & Open Space Rules
Maharashtra (Mumbai)	MCGM / BMC	DCPR 2034, Regulation 30, Table 12; Reg 33.	Island: 1.33 Suburbs: 1.00.	Up to 2.70 (Resi) Up to 5.00 (Comm).	$< 9\text{m}$, $9\text{m} \leq w < 12\text{m}$, $12\text{m} \leq w < 18\text{m}$, $18\text{m} \leq w < 27\text{m}$, $w \geq 27\text{m}$.	Front: 3.0m–9.0m based on road width. Side/Rear: Scale dynamically with building height: $S = 3.0 + \frac{H - 15}{10}$.

State / Region	Planning Authority	Governing Statutory Code & Clauses	Base FSI	Maximum Permissible FSI	Road Width Thresholds	Statutory Setback & Open Space Rules
Maharashtra (Rest of State)	Municipal Corp. (PMC, PCMC, TMC)	UDCPR 2020 (Updated 2025), Tables 6-A & 6-G.	1.10 (Standard Congested: 1.50).	Up to 3.00 (Standard) Up to 4.00 (TOD Corridor).	$< 9\text{m}$, $9\text{m} \leq \text{width} < 12\text{m}$, $12\text{m} \leq \text{width} < 15\text{m}$, $15\text{m} \leq \text{width} < 18\text{m}$, $18\text{m} \leq \text{width} < 24\text{m}$, $\geq 24\text{m}$.	Premium FSI (up to 0.50) + TDR (up to 1.40) loaded based on road width. Height capped by road width: $H \leq 1.5 \times (\text{Road} + \text{Setback})$.
Karnataka (Bengaluru)	BBMP / BDA / BMRDA	Revised Master Plan (RMP 2015 / Draft 2025-2026).	1.75 to 3.25 based on zone.	+40% Premium FAR + 20% TDR loading.	$< 9\text{m}$, $9\text{m} \leq \text{width} < 12\text{m}$, $12\text{m} \leq \text{width} < 15\text{m}$, $15\text{m} \leq \text{width} < 18\text{m}$, $\geq 18\text{m}$.	Premium FAR purchased at 50% guidance value. Front setback scales with road width (Table 8). Side setbacks: min 1.5m to 8.0m.
Delhi NCR	DDA	Master Plan for Delhi (MPD 2021 / Draft 2041).	2.00 (Standard Group Housing).	Up to 3.00 with Purchasable FAR.	Frontage $\geq 18\text{m}$; min plot $3,000\text{m}^2$.	Ground coverage capped at 33.3%. Setbacks: Front 9.0m, Rear 6.0m, Sides 6.0m. Height restricted by aircraft clearance surfaces.
Telangana (Hyderabad)	GHMC / HMDA	Telangana Building	Dynamic / Non-FSI	Uncapped FSI (Density	$< 9\text{m}$, $9\text{m} \leq \text{width} < 12\text{m}$	Height governs

State / Region	Planning Authority	Governing Statutory Code & Clauses	Base FSI	Maximum Permissible FSI	Road Width Thresholds	Statutory Setback & Open Space Rules
		15 & 16.			≥ 9.0 , $9.0 < \text{width} < 12.0$, ≥ 12.0	requires road width ≥ 12 . Front setback min 3.0m–6.0m; rear setback min 3.0m.

Declarative Rule DSL Specification

The platform encodes municipal regulations in a declarative JSON Domain Specific Language:

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "jurisdiction_id": "IN-MH-MUM",
  "authority": "Municipal Corporation of Greater Mumbai",
  "regulation_code": "DCPR-2034",
  "version_tag": "2034.4.2",
  "effective_from": "2018-11-13",
  "effective_to": null,
  "citation": {
    "document": "Development Control and Promotion Regulations for Greater Mumbai, 2034",
    "clause": "Regulation 30, Table 12",
    "gazette_notification": "No. TPB 4317/629/CR-118/2017/UD-11"
  },
  "applicability": {
    "zones": ["RESIDENTIAL_R2", "COMMERCIAL_C1"],
    "min_plot_area_sqm": 0,
    "max_plot_area_sqm": null
  },
  "road_width_rules": [
    {
      "min_road_width_m": 0.0,
      "max_road_width_m": 9.0,
      "base_fsi": 1.00,
      "premium_fsi_max": 0.00,
      "tdr_fsi_max": 0.00,
      "max_building_potential": 1.00,
      "max_permissible_height_m": 15.0,
      "mandatory_front_setback_m": 3.0
    },
    {
      "min_road_width_m": 9.0,
      "max_road_width_m": 12.0,
```

```
"base_fsi": 1.00,
"premium_fsi_max": 0.50,
"tdr_fsi_max": 0.50,
"max_building_potential": 2.00,
"max_permissible_height_m": 32.0,
"mandatory_front_setback_m": 4.5
},
{
  "min_road_width_m": 12.0,
  "max_road_width_m": 18.0,
  "base_fsi": 1.00,
  "premium_fsi_max": 0.50,
  "tdr_fsi_max": 0.70,
  "max_building_potential": 2.20,
  "max_permissible_height_m": 50.0,
  "mandatory_front_setback_m": 6.0
},
{
  "min_road_width_m": 18.0,
  "max_road_width_m": 27.0,
  "base_fsi": 1.00,
  "premium_fsi_max": 0.50,
  "tdr_fsi_max": 0.90,
  "max_building_potential": 2.40,
  "max_permissible_height_m": 70.0,
  "mandatory_front_setback_m": 6.0
},
{
  "min_road_width_m": 27.0,
  "max_road_width_m": null,
  "base_fsi": 1.00,
  "premium_fsi_max": 0.50,
  "tdr_fsi_max": 1.20,
  "max_building_potential": 2.70,
  "max_permissible_height_m": 120.0,
  "mandatory_front_setback_m": 9.0
}
],
"fire_tender_access": {
  "height_threshold_m": 15.0,
  "min_driveway_clearance_m": 6.0,
  "min_turning_radius_m": 9.0,
  "max_dead_end_length_m": 45.0
}
}
```

Deterministic Feasibility Computation Pipeline

The feasibility engine resolves calculations sequentially through a deterministic dependency pipeline.

First, the input parcel boundary in WGS84 (EPSG:4326) is transformed into local metric Cartesian coordinates (EPSG:UTM Zone).

Second, road widening reservations are subtracted along fronting road segments to derive the Net Plot Area (NPA):

$$\text{NPA} = \text{Gross Plot Area (GPA)} - A_{\text{road_widening}} - A_{\text{amenity_reservation}}$$

Surrendered road-widening land generates Transferable Development Rights (TDR) or compensatory FSI under municipal codes.

Third, the declarative regulation engine evaluates the abutting road width against the statutory ruleset to determine base FSI, permissible premium FSI, TDR loading caps, and statutory height limits.

Fourth, the geometric setback engine executes CGAL straight skeleton offsetting along parcel perimeters, applying distinct setback distances to front, rear, and side edges.

Fifth, non-developable zones (utility easements, water body buffers, high-tension electrical clearances) and preserved tree root buffers (2.5m to 4.0m radius) are subtracted using planar boolean difference operations:

$$P_{\text{envelope}} = P_{\text{site}} \setminus \left(\bigcup_i \text{SetbackHalfPlane}_i \cup \bigcup_j \text{Easement}_j \cup \bigcup_k \text{TreeBuffer}_k \right)$$

Sixth, the permissible Built-Up Area (BUA) is computed from the Net Plot Area and FSI potential:

$$\text{Permissible BUA} = \text{NPA} \times (\text{Base FSI} + \text{Premium FSI} + \text{TDR FSI}) + A_{\text{compensatory_fsi}}$$

Seventh, the 3D buildable volume is constructed by extruding the 2D buildable envelope up to the statutory height cap:

$$\text{Max Floors} = \min \left(\left\lfloor \frac{H_{\text{max}} - H_{\text{stilt}}}{H_{\text{floor_to_floor}}} \right\rfloor, \left\lceil \frac{\text{Permissible BUA}}{A_{\text{target_plate}}} \right\rceil \right)$$

Generative Site Planning and Floor-Plate Architecture

When the 3D buildable volume is established, the generative site planning engine places structural building footprints, podiums, and vehicular circulation loops.

Architectural Floor-Plate Dimensional Standards (NBC 2016)

Generated floor plates enforce statutory dimensional minimums based on the National Building Code of India (NBC 2016 Part 3 & Part 4) to ensure buildability and code compliance:

- *Habitable Rooms (Living / Bedrooms)*: Minimum clear floor area of 9.5m² with a minimum horizontal dimension of 2.4m. In multi-room units, secondary bedrooms must measure at least 7.5m² with a minimum width of 2.1m. Clear ceiling height must be at least 2.75m.
- *Kitchens*: Minimum clear floor area of 5.0m² with a minimum width of 1.8m (7.5m² and 2.1m if dining is combined).
- *Bathrooms & Water Closets (W.C.)*: Standalone W.C.: minimum 1.1m² (minimum

width 0.9 m); standalone bathroom: minimum 1.8 m^2 (minimum width 1.2 m); combined toilet-bathroom: minimum 2.8 m^2 (minimum width 1.2 m)).

- *Corridors and Egress Travel Limits*: Minimum residential corridor clear width is 1.5 m (2.0 m for commercial occupancies). High-rise buildings (height $> 15.0\text{ m}$) require a minimum of two enclosed, fire-rated staircases with minimum flight widths of 1.5 m and positive-pressure air shafts. The maximum travel distance from any apartment entry door to the nearest fire exit staircase is 22.5 m , and dead-end corridor lengths cannot exceed 6.0 m .

Floor-Plate Synthesis Stages

Floor-plate generation proceeds through six structured stages:

1. Ingest the 2D buildable envelope derived from setbacks, easements, and tree deductions.
2. Generate structural column grid lines spaced at regular intervals (6.0 m to 7.5 m).
3. Inscribe the vertical circulation core (elevators, pressurized fire stairs, MEP duct risers).
4. Route the central circulation corridor backbone (minimum width 1.5 m , dead-ends $\leq 6.0\text{ m}$).
5. Partition demising walls along column lines to bound individual unit volumes.
6. Pack habitable rooms, kitchens, bathrooms, and wet-service stacks adhering to NBC 2016 norms.

Unit Packing Optimization Algorithm

Packing residential units within an irregular floor plate is formulated as a 2D Constrained Mixed-Integer Linear Program (MILP).

Let the floor plate be discretized into a planar grid of spatial cells $G = \{g_1, g_2, \dots, g_m\}$. Let $U = \{u_1, u_2, \dots, u_k\}$ represent the set of candidate apartment units, where each unit u_j belongs to a typology $T(u_j) \in \{\text{1BHK}, \text{2BHK}, \text{3BHK}\}$ with target carpet area $A_{\text{target}}(T)$ and dimensional tolerance δ .

Objective Function

The objective maximizes saleable carpet area and daylight exposure while minimizing circulation and structural transfer costs:

$$\max Z = w_1 \sum_j \text{CarpetArea}(u_j) - w_2 \text{Area}(\text{Corridors}) - w_3 \text{Cost}(\text{StructuralTransfers}) + w_4 \sum_j \text{DaylightFactor}(u_j)$$

Mathematical Constraints

- *Non-Overlapping Constraint*: Units must not overlap in planar space: $\forall u_a, u_b \in U$ ($a \neq b$), $\text{Polygon}(u_a) \cap \text{Polygon}(u_b) = \emptyset$
- *Corridor Egress Adjacency*: Every unit must share a boundary edge with the circulation corridor: $\forall u_j \in U$, $\partial \text{Polygon}(u_j) \cap \partial \text{Corridor} \geq 1.0\text{ m}$
- *Exterior Daylight Adjacency*: Every habitable unit must share a minimum frontage with the

exterior building envelope for natural ventilation: $\forall u_j \in U, \quad \partial \text{Polygon}(u_j) \cap \partial \text{FloorPlate}_{\{\text{exterior}\}} \geq 3.0 \text{ m}$

- *Vertical Wet-Stack Alignment:* Wet areas (bathrooms and kitchens) must align vertically between adjacent floors within a 5% area tolerance to avoid horizontal plumbing offsets: $\forall \text{Toilet } t \text{ at floor } f, \quad \text{Projection}_{2\text{D}}(t_f) \cap \text{Projection}_{2\text{D}}(t_{f-1}) \geq 0.95 \times \text{Area}(t_f)$

Parking Engine and Structural Grid Integration

The mobility engine proceduralizes the layout of vehicular stalls and drive aisles across subterranean basements, ground stilts, and above-ground podiums. Basements require deep excavation, diaphragm shoring, and dewatering, increasing structural costs. Podiums are faster to construct but consume vertical height allowances.

Vehicle Stall and Mobility Standards

- *Standard Stall Dimensions:* Minimum 2.5 m (width) $\times 5.0 \text{ m}$ (length).
- *Driveway Aisles:* Minimum clear width of 6.0 m for two-way aisles (3.5 m for one-way aisles).
- *Vehicular Ramps:* Maximum gradient of 1:8 (12.5%) for straight ramps (1:10 for curved ramps), with 3.0 m transition zones at top and bottom.
- *Fire Tender Clearance:* Driveways must support an axle load of 45 metric tons, maintaining a continuous clear width of 6.0 m and an inside turning radius of 9.0 m .
- *Electric Vehicle (EV) Allocation:* Minimum 20% of stalls pre-wired for Level 2 EV charging infrastructure.

Structural Bay Grid and Transfer Detection

The structural grid engine spaces reinforced concrete columns on regular 6.0 m to 7.5 m modules. A 7.5 m clear bay accommodates three standard 2.5 m vehicular parking stalls between column faces. When residential unit demising walls above do not align with parking column lines below, the engine flags a structural clash and models a 1.2 m to 1.5 m thick concrete transfer slab, automatically injecting the associated cost premium into the financial pro-forma.

WebGL 3D Massing and BIM/CAD Interoperability

The 3D Massing Engine extrudes 2D floor plates into interactive 3D building meshes rendered directly within the map canvas.

WebGL Lifecycle and Custom Three.js Mapbox Layer

To prevent `CONTEXT_LOST_WEBGL` exceptions caused by creating multiple WebGL canvases, the platform injects a single Three.js scene directly into Mapbox GL JS's existing WebGL rendering loop using a custom layer interface:

```
import mapboxgl from 'mapbox-gl';
```

```

import * as THREE from 'three';

export class MassingThreeLayer implements mapboxgl.CustomLayerInterface {
  id = 'building-massing-three-layer';
  type = 'custom' as const;
  renderingMode = '3d' as const;
  private camera = new THREE.Camera();
  private scene = new THREE.Scene();
  private renderer: THREE.WebGLRenderer | null = null;
  private map: mapboxgl.Map | null = null;

  constructor(private massingMeshes: THREE.Mesh[]) {}

  onAdd(map: mapboxgl.Map, gl: WebGLRenderingContext) {
    this.map = map;
    this.renderer = new THREE.WebGLRenderer({
      canvas: map.getCanvas(),
      context: gl,
      antialias: true
    });
    this.renderer.autoClear = false;

    const ambientLight = new THREE.AmbientLight(0xffffff, 0.8);
    this.scene.add(ambientLight);

    const dirLight = new THREE.DirectionalLight(0xffffff, 0.6);
    dirLight.position.set(50, 100, 50);
    this.scene.add(dirLight);

    this.massingMeshes.forEach(mesh => this.scene.add(mesh));
  }

  render(gl: WebGLRenderingContext, matrix: number[]) {
    const m = new THREE.Matrix4().fromArray(matrix);
    this.camera.projectionMatrix = m;
    this.renderer?.resetState();
    this.renderer?.render(this.scene, this.camera);
    this.map?.triggerRepaint();
  }

  onRemove() {
    this.scene.traverse(object => {
      if (object instanceof THREE.Mesh) {
        object.geometry.dispose();
        if (Array.isArray(object.material)) {
          object.material.forEach(mat => mat.dispose());
        } else {
          object.material.dispose();
        }
      }
    });
  }
}

```

```

    }
  }
});
this.renderer?.dispose();
this.renderer = null;
}
}

```

BIM and CAD Interoperability Pipelines

The platform exports vector drawings and semantic 3D BIM models to ensure interoperability with external architectural software:

- *AutoCAD DXF*: 2D vector layouts are generated via ezdxf, organized into standard CAD layers: A-BNDY-SITE (Site boundary), A-BNDY-SETB (Setbacks), A-BLD-FOOT (Building footprints), A-WALL-EXTR (Demising walls), and A-PARK-STAL (Parking stalls).
- *Industry Foundation Classes (IFC4)*: The platform generates rich 3D BIM models using IfcOpenShell. Georeferencing is established via IfcProjectedCRS and IfcMapConversion, locking the IFC model to the parcel's local UTM coordinate system. Masses decompose into IfcSite, IfcBuilding, IfcBuildingStorey, IfcSpace (with RERA carpet areas), and extruded IfcSlab and IfcWall structural solids.

Institutional Real Estate Financial Feasibility Engine

The financial engine couples procedural layouts with development pro-forma models reflecting Indian institutional underwriting standards.

Mathematical Pro-Forma Equations

- *Gross Development Value (GDV)*: $\text{GDV} = \sum_{i \in \text{Typologies}} \left(A_{\text{saleable}, i} \times P_{\text{base}, i} \right) + N_{\text{parking}} \times P_{\text{stall}} + R_{\text{commercial_lease}} \times \text{}$
- *Total Development Cost (TDC)*: $\text{TDC} = C_{\text{land}} + C_{\text{statutory}} + C_{\text{hard_construction}} + C_{\text{soft_costs}} + C_{\text{finance}} + C_{\text{contingency}}$ Where:
 - $C_{\text{land}} = \text{Agreed Purchase Price} + \text{Stamp Duty (5\%–7\%)} + \text{Registration Fees}$.
 - $C_{\text{statutory}} = \text{Scrutiny Fees} + \text{Development Charges} + \text{Premium FSI Cost} + \text{TDR Purchases}$.
 - $C_{\text{hard}} = \sum_k (\text{BUA}_k \times \text{Rate}_{\text{construction}, k})$.
 - $C_{\text{finance}} = \sum_{t=1}^T \text{Outstanding Debt}_t \times \left(\frac{r_{\text{annual}}}{12} \right)$.
- *Profitability Metrics*: $\text{Gross Development Profit} = \text{GDV} - \text{TDC}$ $\text{Return on Cost (RoC)} = \frac{\text{Gross Profit}}{\text{TDC}}$, $\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{GDV}}$
- *Net Present Value (NPV)*: $\text{NPV} = \sum_{t=0}^T \frac{\text{Net Cash Flow}_t}{(1 + r)^t}$

- *Internal Rate of Return (IRR)*: $\sum_{t=0}^T \frac{\text{Net Cash Flow}_t}{(1 + \text{IRR})^t} = 0$
- *Exact Day-Count Internal Rate of Return (XIRR)*: $\sum_{i=1}^N \frac{\text{Net Cash Flow}_i}{(1 + \text{XIRR})^{\frac{d_i - d_1}{365}}} = 0$
- *Residual Land Value (RLV)*: $\text{RLV} = \frac{\text{GDV}}{\text{statutory}} + C_{\text{hard}} + C_{\text{soft}} + C_{\text{finance}} + C_{\text{contingency}} \{1 + M_{\text{target_profit}}\} - C_{\text{stamp_duty}}$

Scenario Comparison and Multi-Objective Optimization

The platform supports parallel scenario modeling, allowing acquisition teams to evaluate alternative design branches across varying unit mixes, FSI loading assumptions, and construction timelines.

To automate scenario exploration, the system deploys a Multi-Objective Evolutionary Algorithm (NSGA-II). NSGA-II evaluates trade-offs across non-linear objectives without requiring arbitrary weighting factors, producing a Pareto frontier of non-dominated solutions.

Genetic Representation and Objective Functions

A candidate development scenario is encoded as a chromosome vector:

$\mathbf{x} = \langle F_{\text{target}}, R_{\text{resi:comm}}, H_{\text{podium}}, \text{TypologyMix}_{\text{1B:2B:3B}}, \text{GridSpacing}_x, \text{GridSpacing}_y \rangle$

The optimization simultaneously evaluates four objective functions:

1. Maximize Gross Profit: $\max f_1(\mathbf{x}) = \text{GDV}(\mathbf{x}) - \text{TDC}(\mathbf{x})$
2. Maximize Project IRR: $\max f_2(\mathbf{x}) = \text{IRR}(\mathbf{x})$
3. Minimize Structural Complexity: $\min f_3(\mathbf{x}) = \text{Area}(\text{TransferSlabs})$
4. Maximize Daylight Exposure: $\max f_4(\mathbf{x}) = \text{DaylightFactor}(\mathbf{x})$

Statutory violations—such as building heights exceeding road-width caps or parking counts falling below municipal quotas—incur an exterior death-penalty, driving fitness to $-\infty$ to ensure that the Pareto frontier contains only legally buildable designs.

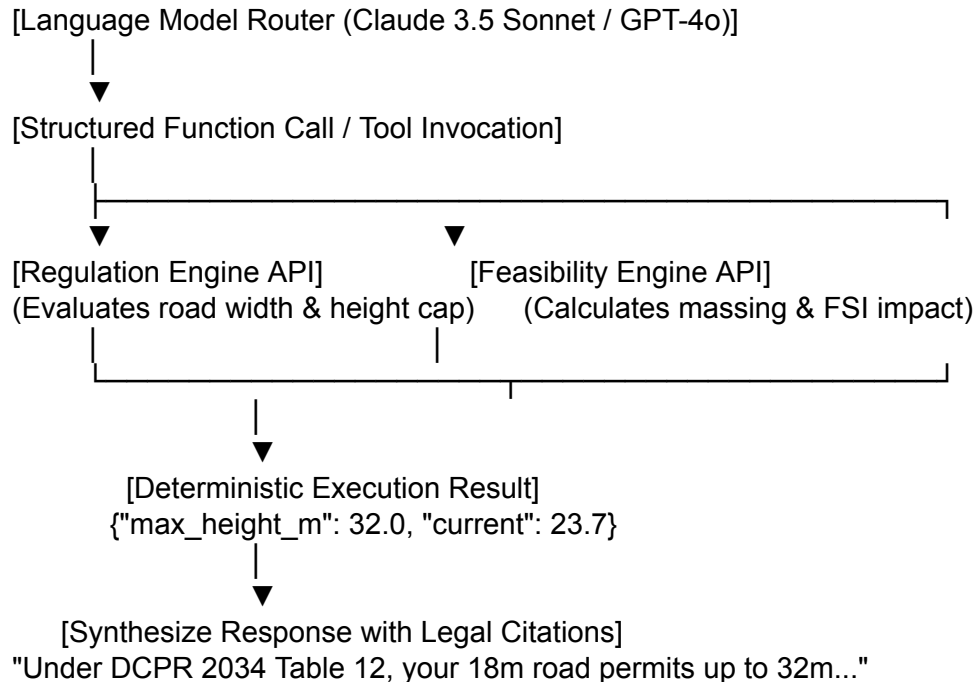
AI Architecture, GraphRAG, and Document Intelligence

The platform strictly separates probabilistic AI inferences from deterministic computational math. Generative design systems fail when they rely on Large Language Models to calculate geometric setbacks or evaluate financial returns, as LLMs are prone to hallucinations and cannot guarantee topological consistency.

Deterministic engines execute all spatial math, geometric transformations, and financial formulas. The AI copilot is restricted to natural-language query routing, summarizing planning codes, and explaining regulatory results.

User Query ("Can I add 2 more floors under Mumbai DCPR 2034?")





Statutory GraphRAG Architecture

Municipal planning gazettes are structured hierarchically: a general setback rule in a chapter may be modified by an exception in a footnote, which is subsequently amended by an administrative circular. Simple text-chunk vector search misses these legal dependencies. The platform deploys a hybrid GraphRAG architecture combining vector search with a property graph database (Neo4j):

- Planning documents are structured into nodes: Authority, RegulationCode, Chapter, Table, Clause, and Exception.
- Graph edges model legal relationships: GOVERNS, DEPENDS_ON, EXEMPTS, and SUPERSEDES.
- Query resolution combines semantic vector search (Qdrant) with graph traversal to retrieve all modifying exceptions and amendments, returning the plain-text answer alongside an auditable legal citation (e.g., [Source: Maharashtra UDCPR 2020, Chapter 6, Table 6-G, Clause 6.3(b)]).

Document Intelligence Pipeline

The document intelligence pipeline ingests scanned revenue surveys and municipal gazettes:

1. Layout analysis models (LayoutLMv3) segment incoming documents into text blocks, tabular dimension schedules, and boundary diagrams.
2. Tabular bye-law schedules are parsed into structured JSON schemas, requiring validation by a licensed municipal architect before being published to the production rule engine.
3. Scanned cadastral survey blueprints are binarized, skeletonized, and vectorized into GeoJSON boundaries, with OCR extracting survey parcel numbers.

API Architecture and Endpoint Specifications

The platform exposes RESTful endpoints supporting JSON serialization, Bearer token authentication, rate limiting, and idempotency headers.

POST /api/v1/auth/login -> Authenticates credentials; issues HttpOnly cookies
GET /api/v1/projects -> Retrieves paginated project records for dashboard
POST /api/v1/projects -> Initializes a new development parcel project
GET /api/v1/projects/:id -> Fetches full project context, geometries, and rules
PUT /api/v1/projects/:id/geometries -> Debounced auto-save endpoint for spatial vector layers
POST /api/v1/projects/:id/feasibility -> Triggers synchronous deterministic feasibility run
POST /api/v1/jobs/generative-layout -> Enqueues asynchronous MILP floor plate synthesis job
GET /api/v1/jobs/:jobId -> Polls status and execution progress of worker jobs
POST /api/v1/projects/:id/scenarios -> Creates or branches a development scenario
POST /api/v1/optimization/nsga2 -> Enqueues multi-objective genetic optimization run
GET /api/v1/reports/:id/feasibility.pdf -> Streams compiled institutional PDF investment prospectus
WS /ws/projects/:id/collaborate -> Bidirectional WebSocket channel for live CRDT edits

Auto-Save Endpoint Contract (PUT /api/v1/projects/:id/geometries)

- **Request Headers:** Content-Type: application/json X-Idempotency-Key: 7b3f9c2a-8924-4f11-9e12-32b491a0c7e2
- **Request Payload:**

```
{
  "client_mutation_timestamp": "2025-05-15T10:30:00.120Z",
  "project_version": 4,
  "features": [
    {
      "type": "Feature",
      "id": "e4b2a3c1-0001-4f11-8e12-000000000001",
      "geometry": {
        "type": "Polygon",
        "coordinates": [[[72.850000, 19.115000], [72.850380, 19.115000], [72.850380, 19.115226], [72.850000, 19.115226], [72.850000, 19.115000]]]]
      },
      "properties": { "entity_type": "SITE_BOUNDARY" }
    }
  ]
}
```

- **Response Status:** 200 OK
- **Response Payload:**

```
{
```

```
"project_id": "b2f6c52a-9999-4c11-8e12-111111111111",
"project_version": 5,
"is_topologically_valid": true,
"gross_plot_area_sqm": 929.03,
"derived_utm_srid": 32643,
"updated_at": "2025-05-15T10:30:00.450Z"
}
```

Frontend Architecture and Map Editor Workspace

The frontend is implemented as a Single Page Application using React 19, TypeScript, TanStack Query, Zustand, and MapLibre GL JS.

The UI isolates the MapLibre map instance and drawing plugins inside a React useRef rather than component state. Storing the map instance in standard component state causes re-renders to re-instantiate the canvas, which exhausts browser WebGL contexts and triggers CONTEXT_LOST_WEBGL crashes. Ephemeral UI state (active tool modes, modal visibility) is managed in Zustand, while server-synchronized data is managed via TanStack Query.

Precision CAD Map Editor Features

- *Object Snapping*: Automatically snaps user input to existing features within a 10px screen radius, supporting vertex, midpoint, edge perpendicular, and 45° / 90° orthogonal angle tracking.
- *Dimension Overlays*: Displays dynamic segment lengths during drafting, allowing users to enter precise numeric lengths.
- *Topology Preservation*: Every vertex adjustment passes through client-side validation checks before dispatch, flagging self-intersections and auto-closing rings.

Security Architecture, Multi-Tenant Isolation, and Auditing

Real estate land feasibility datasets contain sensitive acquisition valuations, joint-venture structures, and bidding pro-formas.

- *Authentication and Token Security*: The platform replaces localStorage token storage with secure, HttpOnly, SameSite=Strict cookies paired with automated refresh token rotation, neutralizing token extraction via Cross-Site Scripting (XSS).
- *Row-Level Security (RLS)*: Multi-tenant data isolation is enforced at the database level via PostgreSQL RLS policies keyed to the active organization ID.
- *Content Security Policy (CSP)*: Enforces strict HTTP CSP headers permitting connections only to authorized tile providers and API gateways: default-src 'self'; script-src 'self'; worker-src 'self' blob:; child-src blob:; connect-src 'self' https://api.mapbox.com https://*.tiles.mapbox.com; img-src 'self' data: blob: https://api.mapbox.com;
- *Calculation Reproducibility*: Every calculation run generates an immutable audit record logging engine versions, rule hashes, geometric inputs, and execution timestamps.

Testing Strategy and Golden Dataset Specification

Quality assurance enforces a multi-tier testing pipeline across geometry math, municipal regulations, and end-to-end browser workflows.

Multi-Tier Testing Harness

- 1. *Geometric Unit Testing*: Validates mathematical functions (Shoelace planar area, Haversine, Vincenty distance, spherical excess) against analytical baselines.
- 2. *Property-Based Geometry Testing (Hypothesis)*: Fuzzes geometry engines with thousands of randomly perturbed, self-intersecting, and degenerate polygons to assert core topological invariants (e.g., buildable envelope area $\leq$ gross plot area).
- 3. *Regulatory Integration Tests*: Asserts that municipal bye-law evaluations match published gazette tables across road width thresholds.
- 4. *End-to-End Canvas Testing (Playwright)*: Simulates automated browser sessions: authenticating, searching an Indian parcel, digitizing boundaries, placing tree markers, and asserting that debounced auto-save mutations persist without errors.

Canonical Golden Test Dataset

The test suite evaluates every release against ten canonical plots representing diverse real-world geometric challenges:

Case ID	Site Geometry Typology	Gross Plot Area (m ²)	Key Spatial Constraints Tested	Expected Buildable Envelope (m ²)	Permissible FSI & Height Cap	Parking Compliance
G-01	Simple Rectangle (40\text{m} \times 23.225\text{m})	929.03	Front 4.5m, Rear 3.0m, Sides 3.0m; 2m road widening.	466.65 (Gross) / 452.45 (Net of tree)	Base 1.0, Max 2.40; Height $\leq 32.0\text{m}$.	22 car stalls compliant.
G-02	Irregular Non-Convex Polygon	2,450.00	5 acute reflex vertices; inward straight skeleton splitting.	1,220.40 (Splits into 2 viable footprints).	Base 1.10, Max 2.70; Height $\leq 50.0\text{m}$.	45 basement stalls.
G-03	Corner Plot (Dual Road Access)	1,800.00	Primary road 24m, secondary road 12m; dual front setbacks.	980.50 (High corner FSI loading unlocked).	Base 1.0, Max 2.70; Height $\leq 70.0\text{m}$.	60 podium stalls.
G-04	Narrow Parcel	900.00	Mandatory 3.0m side	360.00 (Flags	Capped by low height;	12 stilt stalls.

Case ID	Site Geometry Typology	Gross Plot Area (m ²)	Key Spatial Constraints Tested	Expected Buildable Envelope (m ²)	Permissible FSI & Height Cap	Parking Compliance
	(15\text{m} \times 60\text{m})		setbacks reduce width to 9.0m.	narrow structural corridor).	single core.	
G-05	Large Greenfield Master Plan	40,468.60	Surrender 10% Amenity Open Space + 15% Internal Roads.	22,250.00 (Multiple discrete building envelopes).	Base 1.50, TOD Max 4.0; Uncapped height.	850 multi-level stalls.
G-06	Preserved Mature Tree Cluster	3,200.00	6 protected banyan trees with 4.0m radial root buffers.	1,410.20 (Complex organic cutout envelope).	Base 1.0, Max 2.20; Height $\leq 45.0\text{m}$.	70 basement stalls.
G-07	Major Road Widening Reservation	1,500.00	6.0m road widening strip cuts 300 m ² off parcel frontage.	650.00 (Validates TDR generation calculation).	Base 1.0, Max 2.40 + Compensatory TDR.	35 basement stalls.
G-08	High-Tension (HT) Line Easement	4,500.00	20m wide electrical transmission corridor bisecting site.	1,850.00 (Disjoint twin tower envelopes).	Base 1.10, Max 2.50; Height $\leq 60.0\text{m}$.	95 basement stalls.
G-09	Steep Topography / Slope Zone	5,000.00	Slopes > 30% designated non-developable; cut/fill math.	2,100.00 (Stepped terrace massing envelope).	Base 0.75 (Eco-sensitive zoning).	40 surface/stilt stalls.
G-10	Tapered Triangular Plot	1,100.00	Apex angle < 30°; non-constructible sharp corners.	320.00 (Truncates acute apex via minimum width).	Base 1.0, Max 1.80; Height $\leq 24.0\text{m}$.	18 stilt stalls.

Deployment Architecture and Cloud Cost Model

The platform runs on an Amazon Web Services (AWS) infrastructure provisioned via Terraform. The presentation bundle is distributed globally via Cloudflare Enterprise CDN. Ingress traffic routes to an AWS Application Load Balancer terminating TLS 1.3, forwarding requests to an Amazon Elastic Kubernetes Service (EKS) cluster running Kubernetes 1.29. The EKS cluster

hosts API gateway pods, application controllers, Python/C++ geometry worker pools, and Celery task executors. Persistence is managed across an Amazon Aurora PostgreSQL 16 cluster with PostGIS extensions, an ElastiCache Redis 7.2 cluster, and an Amazon S3 bucket for binary CAD/IFC drawing exports and compiled PDF reports.

Cloud Cost Projections Across Scale Tiers

The estimated cloud infrastructure costs across growth stages are summarized below.

Infrastructure Tier / Service	MVP Phase (100 Users)	Growth Phase (10,000 Users)	Enterprise Scale (100,000 Users)	Primary Cost Drivers & Optimization Strategies
Relational Database (Aurora PostGIS)	\$180 / month (db.r6g.large)	\$1,200 / month (db.r6g.4xlarge)	\$5,500 / month (Multi-region replica cluster)	Spatial indexing optimization; query caching in Redis to prevent full table scans.
In-Memory Cache (Redis ElastiCache)	\$60 / month (cache.m6g.large)	\$350 / month (cache.r6g.xlarge cluster)	\$1,400 / month (Distributed Redis Cluster)	Key TTL eviction policies; coordinate payload compression.
Core App & Worker Compute (EKS)	\$250 / month (3 nodes, c6i.xlarge)	\$2,400 / month (16 nodes, auto-scaled)	\$12,000 / month (Karpenter auto-scaling spot nodes)	Offload compute to spot worker instances; auto-scale geometry workers based on queue depth.
GPU Compute (Layout & Ingestion)	\$0 / month (Shared CPU fallback)	\$950 / month (2x g5.xlarge instances)	\$4,800 / month (Multi-GPU Ray cluster)	Batch OCR document processing overnight; on-demand GPU spinning.
Object Storage & CDN (S3 + Cloudflare)	\$25 / month (Standard S3)	\$220 / month (S3 Intelligent-Tiering)	\$1,100 / month (Global Cloudflare Enterprise CDN)	Cache static vector tiles at the edge; archive older scenario revisions to Glacier.
AI LLM API & Vector Indexing	\$100 / month (API usage)	\$1,500 / month (Fine-tuned inference + Qdrant)	\$8,000 / month (Dedicated self-hosted vLLM pods)	Cache identical regulatory queries; strict tool routing before prompting LLMs.
Map Basemap & Geocoding APIs	\$50 / month (Mapbox)	\$800 / month (Volume tier)	\$3,500 / month (Self-hosted)	Self-host open-source

Infrastructure Tier Service	MVP Phase (100 Users)	Growth Phase (10,000 Users)	Enterprise Scale (100,000 Users)	Primary Cost Drivers & Optimization Strategies
	pay-as-you-go)		OpenMapTiles tileserver)	vector tileservers via MapLibre at high enterprise scale.
Observability & Logging (Datadog/OTel)	\$80 / month	\$600 / month	\$2,800 / month	Filter high-frequency canvas heartbeat logs; sample non-critical traces.
Total Estimated Cloud Expenditure	\$745 / month	\$8,020 / month	\$39,100 / month	Blended monthly cost per active user drops from \$7.45 down to \$0.39 at scale.

Open-Source Versus Commercial Dependency Strategy

To avoid commercial vendor lock-in while ensuring regulatory compliance, software dependencies are classified across licensing tiers.

Technology Component	Category & Primary License	Commercial Cost Impact	Vendor Lock-in Risk	Replacement / Exit Strategy
PostgreSQL & PostGIS	Open Source (PostgreSQL / GPLv2)	Free / Hosting only.	None. Global industry standard.	Fully portable across all major cloud managed database platforms.
CGAL 6.0 (Straight Skeleton)	Open Source (LGPL / GPLv3)	Free for isolated microservice binaries.	Low. Self-contained C++ shared library.	Can be replaced with Boost.Polygon or custom straight skeleton implementations.
GEOS & Shapely	Open Source (LGPLv2.1 / BSD-3)	Free / Open-source.	None.	Standard C-API; drop-in replacement via Clipper2 or CGAL.
Mapbox GL JS v3	Commercial (Proprietary / Per-Map-Load)	Scales with monthly active users (\$5 per 1,000 loads).	Moderate. Tied to Mapbox basemap styles.	Migrate directly to open-source MapLibre GL JS with self-hosted OpenMapTiles.
Three.js	Open Source	Free /	None.	Babylon.js or

Technology Component	Category & Primary License	Commercial Cost Impact	Vendor Lock-in Risk	Replacement / Exit Strategy
	(MIT)	Open-source.		WebGPU-native implementations.
Google OR-Tools	Open Source (Apache 2.0)	Free / Open-source.	None.	Commercial solvers (Gurobi, CPLEX) or open-source SCIP/HiGHS.
lfcOpenShell	Open Source (LGPLv3)	Free / Open-source.	None.	XBim or commercial Autodesk Construction Cloud APIs.
Claude 3.5 Sonnet / OpenAI	Commercial (Proprietary API)	Pay-per-token API consumption.	High if prompts are coupled to proprietary tool schemas.	Abstract behind LiteLLM to enable drop-in replacement with open-source Llama 3 models.

P0 Minimum Viable Product (MVP) and Implementation Order

The Minimum Viable Product (P0) focuses strictly on delivering the core end-to-end site feasibility loop, postponing district master planning and automated document parsing to later releases.

Feature Release Tiers

- *P0 (MVP - Days 1–90)*: Complete end-to-end feasibility workflow. User creates a project, digitizes a boundary on the map canvas, applies setbacks and site deductions, calculates the net buildable envelope, compiles permissible FSI under local regulations, and exports an institutional PDF feasibility report.
- *P1 (Growth - Months 4–6)*: Generative floor plates, NBC 2016 unit packing (1BHK–3BHK), subterranean and podium parking layouts, and Three.js 3D massing.
- *P2 (Enterprise - Months 7–9)*: Multi-objective genetic optimization (NSGA-II), vector BIM/CAD exports (IFC4, DXF), real-time collaborative map editing via CRDTs, and multi-parcel consolidation.
- *P3 (Advanced Intelligence - Months 10–12)*: Automated raster-to-vector cadastral blueprint ingestion, deep learning table parsing, and GraphRAG over municipal gazettes.

P0 MVP Feature Matrix

Feature Module	Priority Tier	Technical Complexity	Core Dependencies	Deliverable Description
GeoJSON Parcel	P0	Moderate	MapLibre GL JS,	Digitize parcel

Feature Module	Priority Tier	Technical Complexity	Core Dependencies	Deliverable Description
Delineation			Mapbox Draw	boundaries; automatic ring closure and self-intersection prevention.
Metric Coordinate Projection	P0	Moderate	PyProj, PostGIS ST_Transform	Automated conversion of WGS84 coordinates to local metric UTM zone.
Straight Skeleton Setbacks	P0	High	CGAL 6.0, Shapely 2.0	Inward mitered setback offsetting without rounded corners.
DCPR 2034 / UDCPR Encoders	P0	High	Declarative Rule DSL	Evaluate road widths, derive base/premium FSI, and cap permissible building heights.
Site Constraint Subtractions	P0	Moderate	GEOS / PostGIS ST_Difference	Subtract road widening, utility easements, and radial tree buffers from gross area.
Buildable Envelope 3D	P0	Moderate	Three.js Custom WebGL Layer	Extrude 3D building volume representing maximum statutory massing envelope.
Core Financial Pro-Forma	P0	Moderate	Deterministic DCF Engine	Model GDV, TDC, statutory charges, IRR, and Residual Land Value (RLV).
Automated PDF Export	P0	Low	Headless Chromium, PDFKit	Render professional multi-page land feasibility investment memorandum.

90-Day Implementation Roadmap

The engineering roadmap delivers the production P0 MVP across six structured two-week

sprints.

Sprint Schedule and Deliverables

- *Weeks 1–2 (Sprint 1: Architecture, Scaffolding, and Database Schemas)*: Initialize monorepo structure using Turborepo and pnpm workspaces. Deploy PostgreSQL 16 with PostGIS 3.4 extensions on AWS RDS, applying initial DDL schemas. Scaffold the React 19 / TypeScript client application with Vite and Tailwind CSS.
- *Weeks 3–4 (Sprint 2: Authentication, Tenant Context, and Map Workspace)*: Implement authentication controllers with HttpOnly cookie handling and refresh token rotation. Build the MapLibre/Mapbox GL JS workspace, isolating the canvas instance inside a React useRef hook. Implement the 1000ms debounced auto-save hook with AbortController cancellation.
- *Weeks 5–6 (Sprint 3: Computational Geometry Engine)*: Deploy the Python/C++ Geometry Computation Worker container. Implement automated UTM zone projection via PyProj and PostGIS. Integrate CGAL 6.0 straight skeleton inward offsetting to generate sharp, mitered setbacks. Implement multi-layer boolean subtractions for road widenings, easements, and radial tree buffers.
- *Weeks 7–8 (Sprint 4: Declarative Regulation Engine and Feasibility Pipeline)*: Implement the declarative Rule DSL JSON schema and execution engine. Ingest and validate Mumbai DCPR 2034 Table 12 and Maharashtra UDCPR Table 6-A/6-G rule definitions. Build the deterministic Feasibility Pipeline resolving Net Plot Area, Base FSI, Premium FSI, TDR loading, and height caps. Render dynamic HUD metrics and 3D buildable massing envelopes inside the WebGL canvas.
- *Weeks 9–10 (Sprint 5: Financial Feasibility & Scenarios)*: Implement the deterministic real estate pro-forma engine calculating GDV, TDC, approval premiums, and construction cash flows. Implement financial equations for Net Present Value (NPV), annualized IRR, and Residual Land Value (RLV). Build the Scenario Manager, allowing users to branch projects, adjust densities, and compare results.
- *Weeks 11–12 (Sprint 6: Reporting, QA Hardening & Launch)*: Build the automated PDF reporting engine streaming institutional land feasibility prospectuses. Author Playwright end-to-end integration test suites across the ten canonical golden dataset plots. Enforce PostgreSQL Row-Level Security, audit CSP headers, and configure rate limiting. Deploy production cluster on AWS EKS behind Cloudflare; onboard initial cohort of institutional development partners.

Enterprise Monorepo Codebase Architecture

The project is organized as a modular monorepo using Turborepo and pnpm workspaces to enforce architectural separation across domains:

- `/apps/web`: React 19 Single Page Application, TypeScript, MapLibre GL JS, Tailwind CSS, TanStack Query.
- `/apps/api`: Node.js (NestJS) API Gateway, REST endpoints, WebSocket collaboration gateway, session handlers.
- `/services/geometry-worker`: Python 3.11 / C++ service running CGAL 6.0, Shapely 2.0, PyProj, and GEOS for spatial mathematics.
- `/services/regulation-service`: Declarative Rule DSL evaluator and municipal gazette

compiler.

- /services/feasibility-engine: Deterministic calculation pipeline deriving permissible built-up areas and 3D buildable volumes.
- /services/optimization-worker: Google OR-Tools (MILP) unit packing and Platypus (NSGA-II) genetic algorithm solvers.
- /services/financial-service: Institutional DCF pro-forma, cash-flow discounting, and sensitivity calculators.
- /services/report-generator: Headless Chromium and PDFKit export pipeline streaming investment dossiers.
- /packages/domain-models: Canonical TypeScript interfaces, entity schemas, and lifecycle definitions.
- /packages/spatial-contracts: RFC 7946 GeoJSON Zod schemas, coordinate converters, and topology validators.
- /packages/regulatory-rulesets: Versioned JSON DSL rule definitions (DCPR 2034, UDCPR 2025, JBBL 2016).
- /packages/ui-components: Shared React design system components, Radix UI primitives, and modal overlays.
- /infrastructure/docker: Multi-stage Dockerfiles and local docker-compose orchestration environments.
- /infrastructure/kubernetes: Production Helm charts, EKS deployment manifests, ingress routes, and autoscalers.
- /infrastructure/terraform: Infrastructure-as-code configurations for AWS VPC, RDS Aurora, Redis, and EKS clusters.
- /tests/golden-dataset: Regression harness maintaining the ten canonical golden plots with analytical baselines.

Complete Worked Feasibility Walkthrough

To substantiate the integrated operation of the platform's geometric, regulatory, vertical massing, and financial engines, the following trace details an end-to-end feasibility calculation for an actual urban redevelopment parcel.

1. Parcel Input & Site Context

- *Location*: Andheri East, Mumbai Suburban District, Maharashtra, India.
- *Governing Regulation*: Development Control and Promotion Regulations for Greater Mumbai (DCPR 2034, Regulation 30, Table 12).
- *Cadastral Coordinates (WGS84, EPSG:4326)*:
 - v_0 : [72.850000°E, 19.115000°N]
 - v_1 : [72.850380°E, 19.115000°N]
 - v_2 : [72.850380°E, 19.115226°N]
 - v_3 : [72.850000°E, 19.115226°N]
- *Projected Coordinate Reference System*: UTM Zone 43N (EPSG:32643).
- *Parcel Dimensions*: Rectangular geometry measuring 40.00 m (frontage along south) $\times$ 23.225 m (depth).
- *Gross Plot Area (GPA)*: $40.00 \text{ m} \times 23.225 \text{ m} = 929.03 \text{ m}^2$

- $d_{\text{span}} \text{ m}^2 \text{ quad } (10,000.00 \text{ m}^2 \text{ sq ft})$.
- Abutting Access Road Width:** 18.00 m along the southern frontage.
- Statutory Road Widening Reservation:** 2.00 m reservation line along the entire 40.00 m frontage.
- Preserved Mature Tree:** One mature Banyan tree located at projected metric coordinates (30.00 m, 12.00 m) with a mandatory 2.50 m radial root protection buffer.

2. Spatial Deductions & Net Developable Envelope

- Road Widening Deduction Area:** $A_{\text{widening}} = 40.00 \text{ m} \times 2.00 \text{ m} = 80.00 \text{ m}^2 \text{ quad } (861.11 \text{ m}^2 \text{ sq ft})$
- Net Plot Area (NPA):** $\text{NPA} = \text{GPA} - A_{\text{widening}} = 929.03 \text{ m}^2 - 80.00 \text{ m}^2 = 849.03 \text{ m}^2 \text{ quad } (9,138.89 \text{ m}^2 \text{ sq ft})$
- Mandatory Municipal Setbacks (DCPR 2034 Table 12 & NBC 2016):**
 - Front Setback (measured from post-widening road line): 4.50 m.
 - Rear Setback (northern boundary): 3.00 m.
 - Side Setbacks (eastern and western boundaries): 3.00 m each.
- Buildable Footprint Dimensions:** $\text{Buildable Width} = 40.00 \text{ m} - (3.00 \text{ m} + 3.00 \text{ m}) = 34.00 \text{ m}$ $\text{Buildable Depth} = (23.225 \text{ m} - 2.00 \text{ m}) - (4.50 \text{ m} + 3.00 \text{ m}) = 13.725 \text{ m}$ $\text{Gross Footprint Area} = 34.00 \text{ m} \times 13.725 \text{ m} = 466.65 \text{ m}^2 \text{ quad } (5,022.98 \text{ m}^2 \text{ sq ft})$
- Tree Root Buffer Deduction:** The circular root buffer ($\text{Radius} = 2.50 \text{ m}$) overlaps the buildable footprint by 14.20 m^2 . $\text{Net Buildable Footprint Area} = 466.65 \text{ m}^2 - 14.20 \text{ m}^2 = 452.45 \text{ m}^2 \text{ quad } (4,870.13 \text{ m}^2 \text{ sq ft})$

3. Statutory FSI Allocation & Vertical Massing

Under DCPR 2034 Table 12 for an 18.00 m road in suburban Mumbai:

- $\text{Base FSI} = 1.00$
- $\text{Premium FSI} = 0.50$ (purchased at 50% of the Annual Statement of Rates [ASR])
- $\text{TDR Loading} = 0.90$ (purchased on open market)
- $\text{Maximum Permissible FSI} = 1.00 + 0.50 + 0.90 = 2.40$
- Permissible Built-Up Area (BUA):** $\text{Permissible BUA} = \text{NPA} \times \text{FSI} = 849.03 \text{ m}^2 \times 2.40 = 2,037.67 \text{ m}^2 \text{ quad } (21,933.34 \text{ m}^2 \text{ sq ft})$
- Vertical Stacking Configuration:**
 - Ground Level: Stilt parking, entrance lobby, and fire command center (4.20 m floor-to-floor height).
 - Habitable Floors: Floor plate designed at 340.00 m^2 (3,659.73 m² sq ft), well within the 452.45 m² buildable footprint to preserve open space for external services and balconies. $\text{Floor Count} = \frac{2,037.67 \text{ m}^2}{340.00 \text{ m}^2} \approx 6$ Habitable Floors
 - Total Building Height: $H = 4.20 \text{ m}_{\text{stilt}} + (6 \times 3.05 \text{ m}_{\text{resi}}) + 1.20 \text{ m}_{\text{parapet}} = 23.70 \text{ m}$ The 23.70 m height is compliant with the 32.0 m statutory cap for an 18 m

m} road and satisfies NBC high-rise fire safety requirements.

4. Floor Plate Architecture and Unit Mix

Each of the 6 habitable floor plates (340.00 m^2) is laid out to optimize circulation efficiency and natural daylight access:

- **Vertical Circulation Core:** Central structural core with two 8-passenger elevators, two fire-rated staircases (1.50 m flight width), and an MEP utility duct. Total core area: 48.00 m^2 .
- **Central Corridor:** Single-loaded corridor measuring $1.80\text{ m} \times 12.00\text{ m} = 21.60\text{ m}^2$.
- **Net Habitable Carpet Area per Floor:** $340.00\text{ m}^2 - (48.00\text{ m}^2 + 21.60\text{ m}^2) = 270.40\text{ m}^2$ (2,910.56 sq ft)
- **Unit Mix Distribution per Floor:**
 - Two 2BHK units: 75.00 m^2 RERA carpet area each.
 - Two 1BHK units: 45.00 m^2 RERA carpet area each.
 - Common interior partition walls and shafts: 30.40 m^2 .
- **Total Development Program:** $6\text{ floors} \times 4\text{ units/floor} = \mathbf{24\text{ Units}}$ (12 $\times$ 2BHK, 12 $\times$ 1BHK).
- **Total RERA Carpet Area:** $6 \times (2 \times 75.0 + 2 \times 45.0) = \mathbf{1,440.00\text{ m}^2}$ (15,500.03 sq ft).
- **Total Saleable Area (using 38% super-built-up loading factor):** $15,500.03\text{ sq ft} \times 1.38 = \mathbf{21,390.04\text{ sq ft}}$ (1,987.20 m^2)

5. Parking Stall Layout and Compliance

Statutory Parking Norms under Mumbai DCPR 2034:

- 2BHK units: 1 stall per unit $\rightarrow 12\text{ stalls}$.
- 1BHK units: 0.5 stalls per unit $\rightarrow 6\text{ stalls}$.
- Visitor Parking: 10% statutory surcharge $\rightarrow 2\text{ stalls}$.
- **Total Statutory Parking Required:** $\mathbf{20\text{ Car Stalls}}$.

Parking Provision:

- Ground Stilt Level: 8 surface stalls + 10 two-wheeler bays.
- Basement Level: 600.00 m^2 single-level subterranean basement accommodating 14 stalls, a 6.0 m two-way circulation aisle, and a 1:8 access ramp.
- **Total Parking Provided:** $\mathbf{22\text{ Car Stalls}}$ (Compliant, +10% buffer).

6. Financial Pro-Forma Underwriting Summary

All currency values are expressed in Indian Rupees (INR).

Capital Revenue (Gross Development Value - GDV)

- Residential Saleable Area: $21,390\text{ sq ft} @ \text{INR } 22,000 / \text{sq ft} = \text{INR } 47,05,80,000$
- Covered Car Parking Sales: $20\text{ Allocated Stalls} @ \text{INR } 5,00,000 = \text{INR } 1,00,00,000$
- **Gross Development Value (GDV):** $\mathbf{\text{INR } 48,05,80,000}$

Total Development Cost (TDC)

- Land Acquisition Cost ($9,138.89 \text{ sq ft NPA} @ \text{INR } 19,700/\text{sq ft}$): $\text{INR } 18,00,00,000$
- Stamp Duty & Legal Registration Charges (~7%): $\text{INR } 1,26,00,000$
- Premium FSI Purchase ($0.50 \text{ FSI} = 4,569 \text{ sq ft} @ 50\% \text{ ASR}$): $\text{INR } 2,28,45,000$
- TDR Purchase ($0.90 \text{ FSI} = 8,225 \text{ sq ft}$ open market purchase): $\text{INR } 2,87,87,500$
- Municipal Scrutiny & Development Charges: $\text{INR } 85,00,000$
- Basement Construction ($6,458 \text{ sq ft BUA} @ \text{INR } 3,200/\text{sq ft}$): $\text{INR } 2,06,65,600$
- Superstructure Construction ($25,000 \text{ sq ft GFA} @ \text{INR } 2,800/\text{sq ft}$): $\text{INR } 7,00,00,000$
- Architectural, Structural, & Geotechnical Consultancy Fees: $\text{INR } 65,00,000$
- Marketing, Brokerage, & Transaction Commissions (3% of GDV): $\text{INR } 1,44,17,400$
- Construction Finance & Debt Servicing Charges: $\text{INR } 2,40,00,000$
- Project Contingency Reserve (4% of Hard Construction Costs): $\text{INR } 36,26,624$
- **Total Development Cost (TDC): $\text{INR } 39,19,42,124$**

Profitability and Investment Returns

- **Gross Development Profit** ($\text{GDV} - \text{TDC}$): $\text{INR } 8,86,37,876$
- **Developer Return on Cost (RoC):** $\text{RoC} = \frac{\text{INR } 8,86,37,876}{\text{INR } 39,19,42,124} \times 100 = \text{22.61\%}$
- **Gross Profit Margin** ($\text{Profit} / \text{GDV}$): $\text{18.44\%}$
- **Annualized Project IRR (36-Month Development Horizon):** $\text{24.85\%}$
- **Net Present Value (NPV @ 12% Cost of Capital):** $\text{INR } 3,94,20,540$
- **Residual Land Value (RLV @ 20% Target Margin on Cost):** $\text{INR } 18,92,45,000$

7. Sensitivity Analysis Matrix

The sensitivity of project profitability across construction cost and sales realization variances is evaluated below.

Construction Cost Variance \ Sales Realization Variance	Downside Case (-10% Realization: INR 19,800/sqft)	Base Case (INR 22,000/sqft)	Upside Case (+10% Realization: INR 24,200/sqft)
High Construction Cost (+10%)	Profit: INR 3,25,12,714 RoC: 8.11% IRR: 10.4%	Profit: INR 7,95,70,714 RoC: 20.03% IRR: 21.8%	Profit: INR 12,66,28,714 RoC: 31.87% IRR: 32.1%
Base Construction Cost (INR 2,800/sqft)	Profit: INR 4,15,79,876 RoC: 10.61% IRR: 13.2%	Profit: INR 8,86,37,876 RoC: 22.61% IRR: 24.8%	Profit: INR 13,56,95,876 RoC: 34.62% IRR: 35.4%
Low Construction Cost (-10%)	Profit: INR 5,06,47,038 RoC: 13.23% IRR:	Profit: INR 9,77,05,038 RoC: 25.52% IRR:	Profit: INR 14,47,63,038 RoC:

Construction Cost Variance \ Sales Realization Variance	Downside Case (-10% Realization: INR 19,800/sqft)	Base Case (INR 22,000/sqft)	Upside Case (+10% Realization: INR 24,200/sqft)
	16.1%	27.9%	37.81% IRR: 38.6%

Technical Synthesis and System Verification

The engineering architecture specified in this blueprint transitions real estate land acquisition from fragmented, weeks-long manual drafting into an automated, explainable, and reproducible software pipeline. By replacing generic Web Mercator approximations with local metric UTM projections, and exchanging rounded Minkowski buffering for exact CGAL straight skeleton algorithms, the platform ensures spatial and legal correctness.

By decoupling probabilistic AI models into pure summarization and natural-language assistance roles—while reserving all geometric boolean clipping, regulatory bye-law compilations, and financial cash flow models for deterministic execution engines—the platform guarantees absolute calculation reproducibility and statutory auditability. The system enables real estate developers, urban master planners, institutional investors, and municipal authorities to transform raw land parcels into optimized, fully underwritten, and legally compliant development configurations within minutes.

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